



今日的語言模型 是如何做文字接龍的 Transformer 簡介

前言

這門課是生成式 AI 導

語言模型

論

找出參數

輸入：人工智

輸出：慧

輸入：不要忘了今天來開

輸出：會

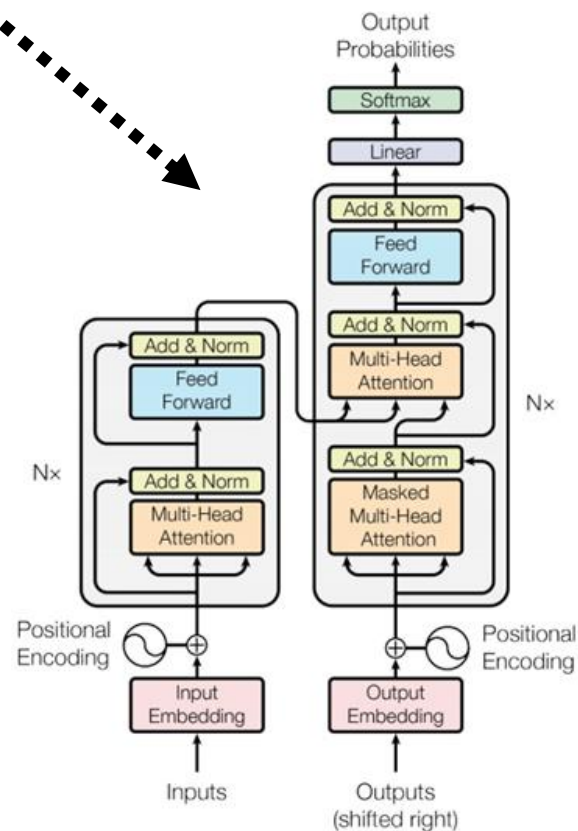
輸入：床前明月

輸出：光

⋮

⋮

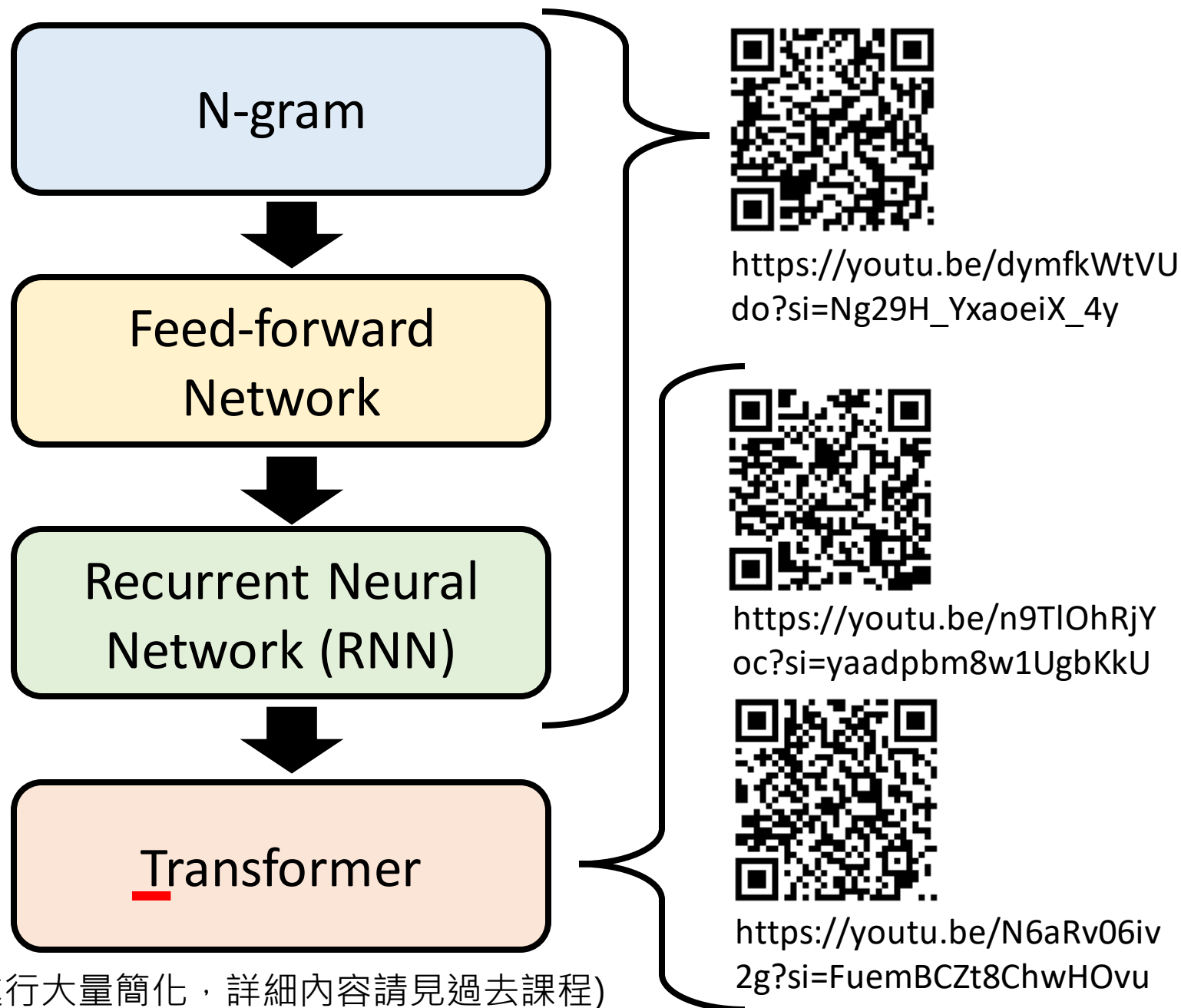
訓練資料



類神經網路

Transformer

模型演進

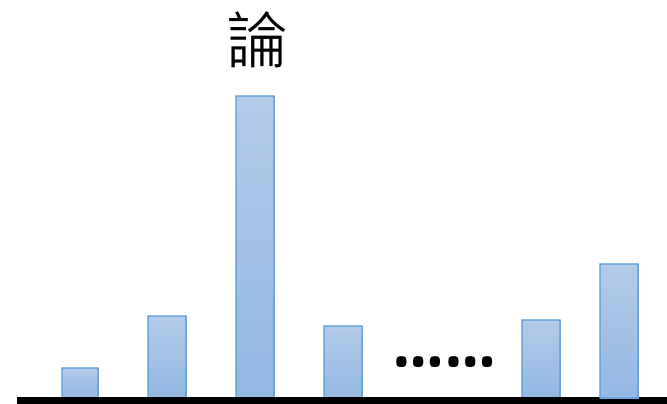


(本課程會對 Transformer 的說明進行大量簡化，詳細內容請見過去課程)

Transformer 概述

這門課是生成式 AI 導

語言模型



1.Tokenization

文字轉為 Token

2. Input
Layer

理解Token

3. Attention

理解上下文

4. Feed
Forward

整合、思考 ...

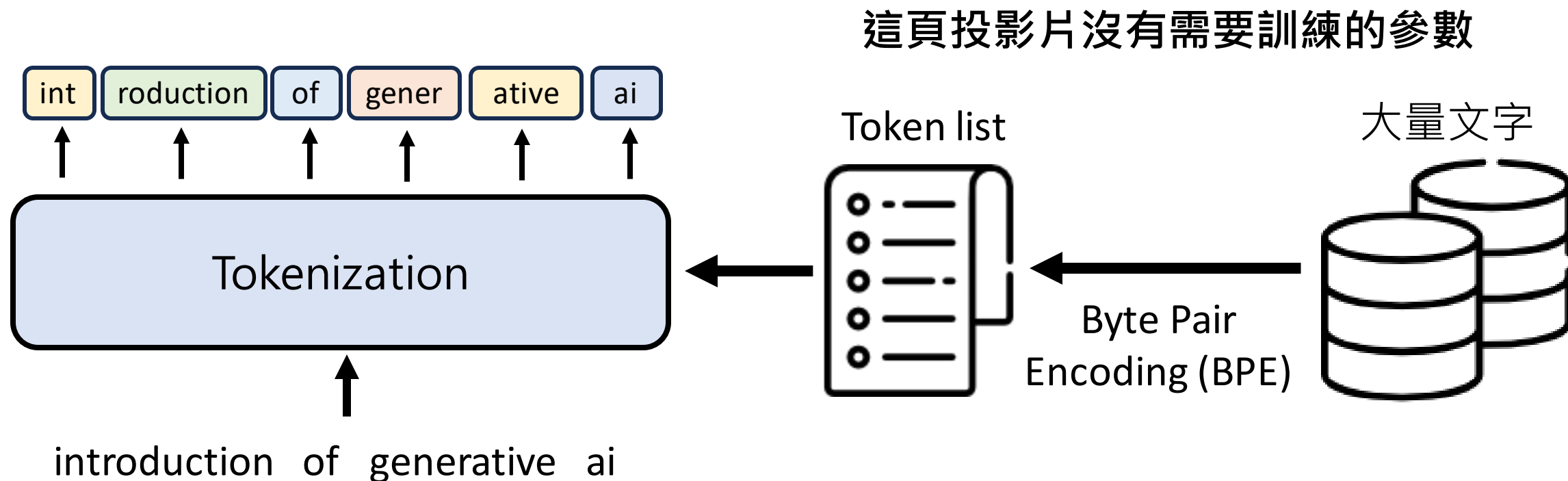
5. Output
Layer

得到輸出

Transformer Block X N 反覆思考

1. 把文字變成 Token

- 語言模型是以 Token 作為單位來對文字進行處理



GPT-3.5 & GPT-4 GPT-3 (Legacy)

A language model is a probabilistic model of a natural language. In 1980, the first significant statistical language model was proposed, and during the decade IBM performed 'Shannon-style' experiments, in which potential sources for language modeling improvement were identified by observing and analyzing the performance of human subjects in predicting or correcting text.



Tokens

65

Characters

373

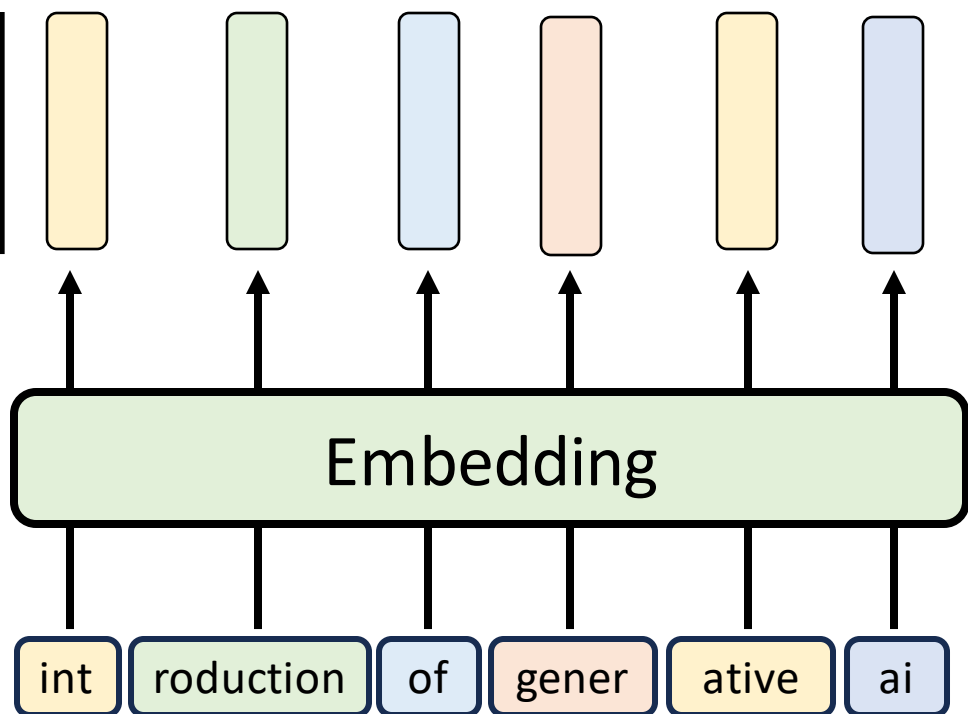
A language model is a probabilistic model of a natural language. In 1980, the first significant statistical language model was proposed, and during the decade IBM performed 'Shannon-style' experiments, in which potential sources for language modeling improvement were identified by observing and analyzing the performance of human subjects in predicting or correcting text.

Text Token IDs

<https://platform.openai.com/tokenizer>

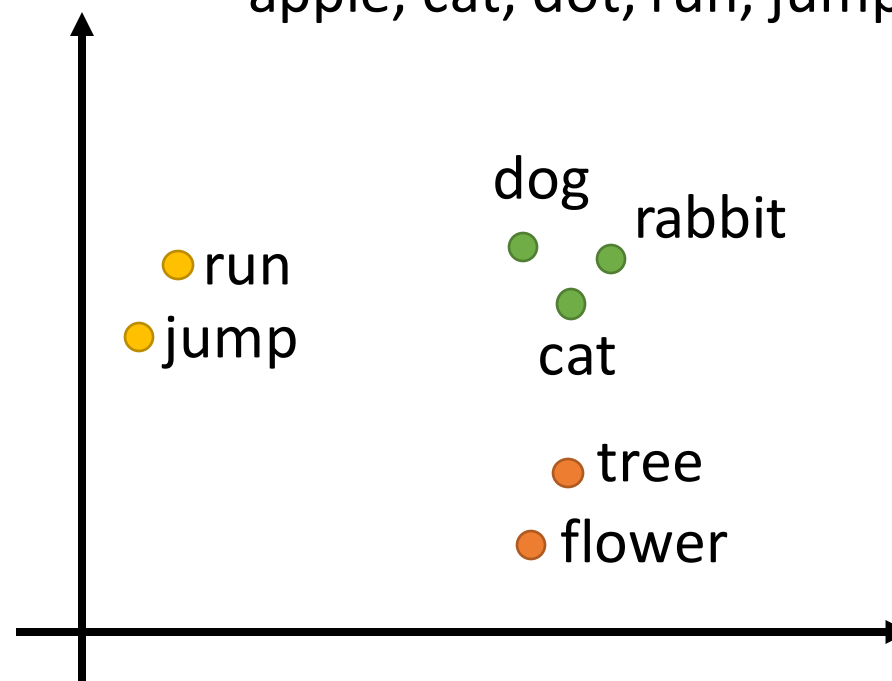
2. 理解每個 Token — 語意

向量 (Vector)

$$\begin{bmatrix} 1.2 \\ -3.5 \\ \vdots \\ -1.9 \\ 7.2 \end{bmatrix}$$


原本每一個 Token 都是獨立的符號

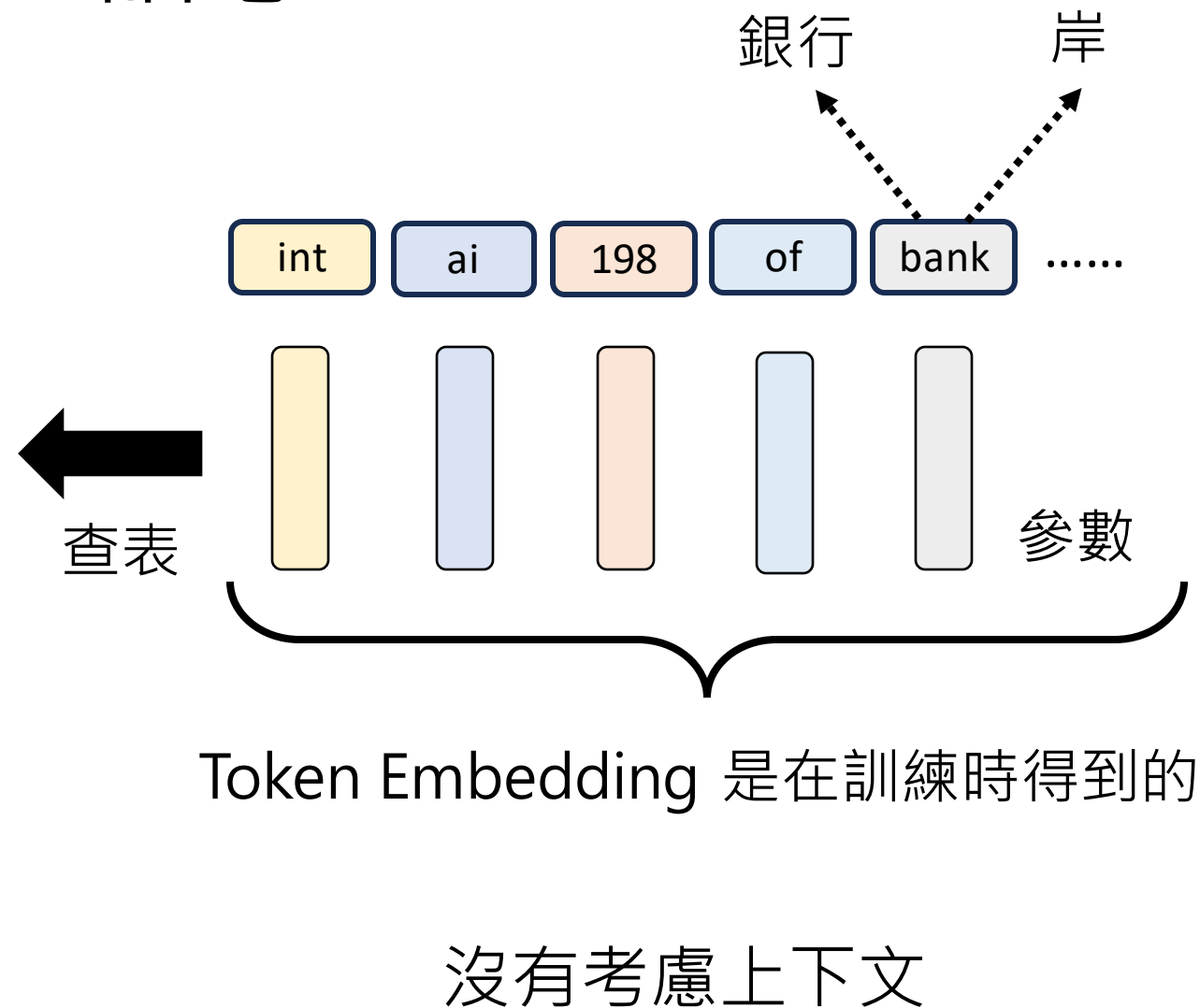
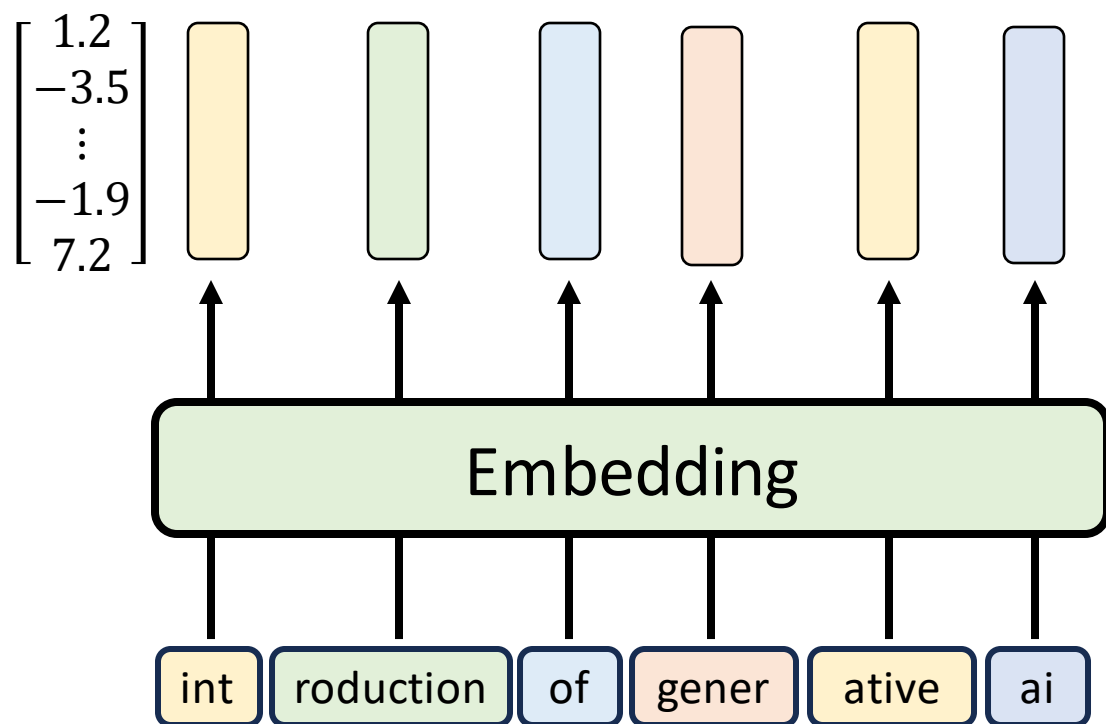
apple, cat, dot, run, jump



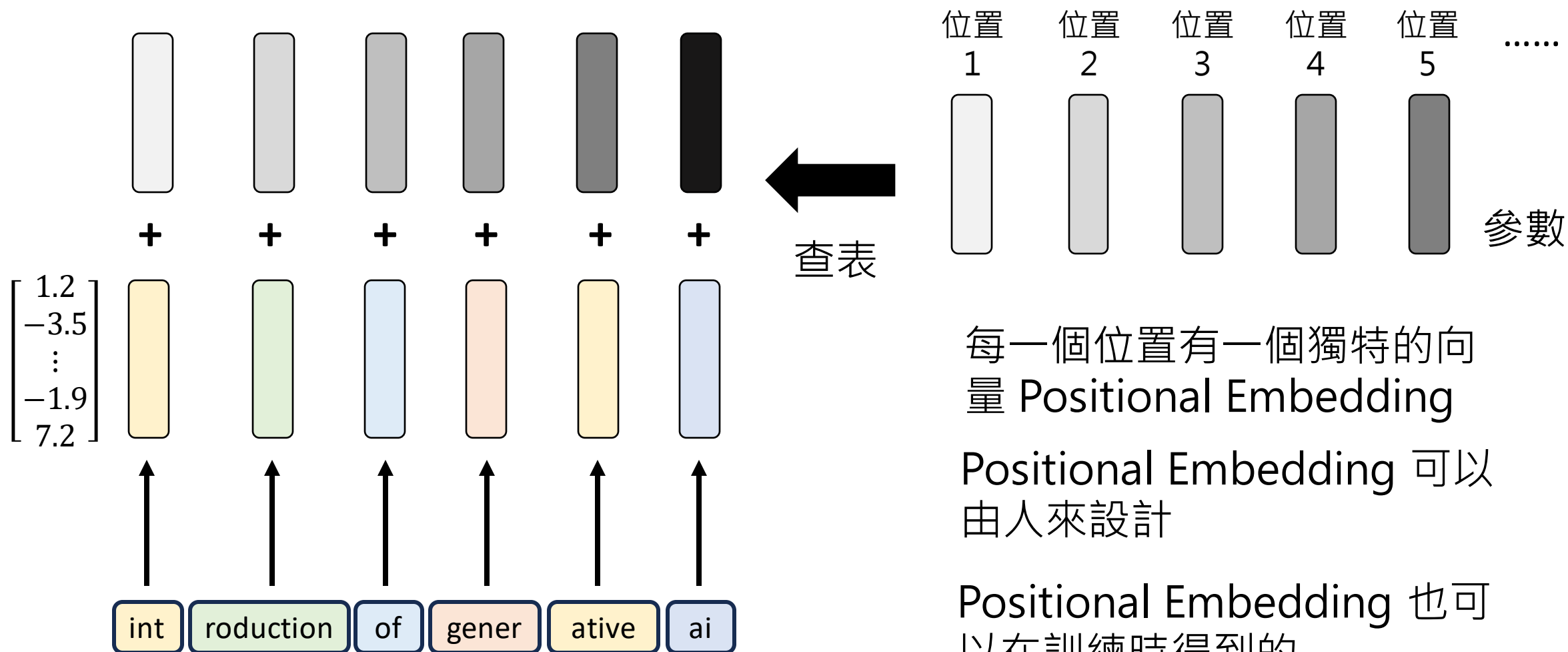
意思相近的 Token 會有接近的 Embedding

2. 理解每個 Token — 語意

意思相近的 Token 會有接近的 Embedding

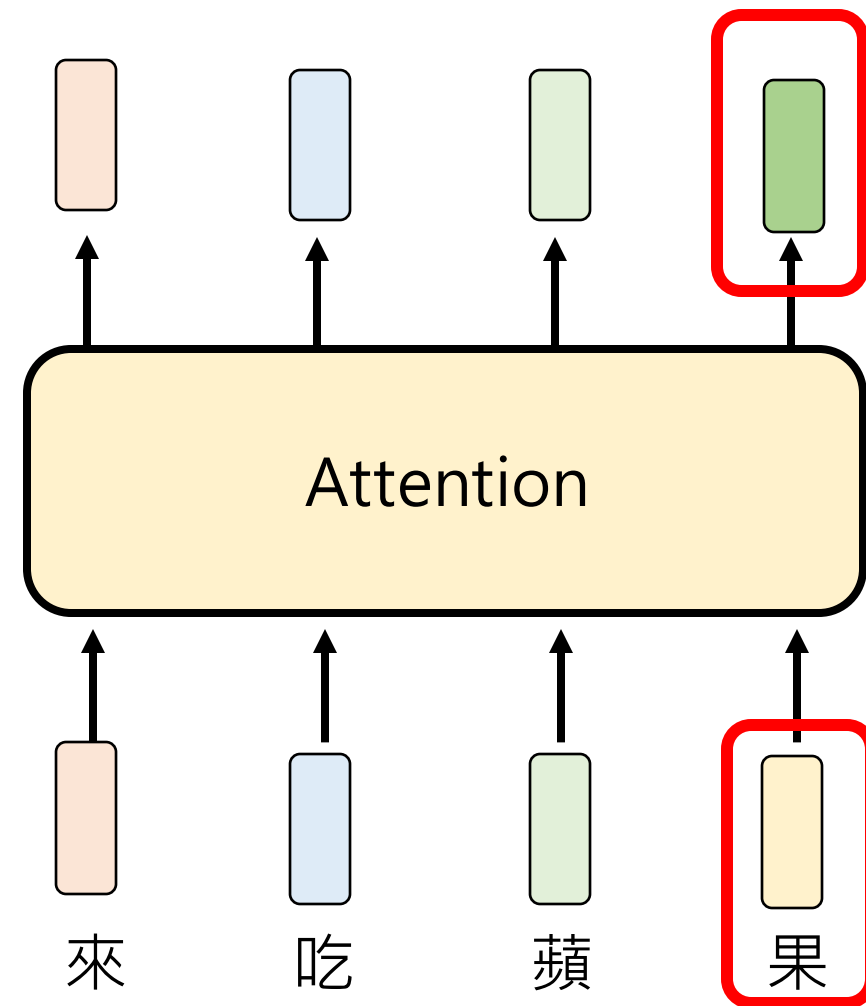
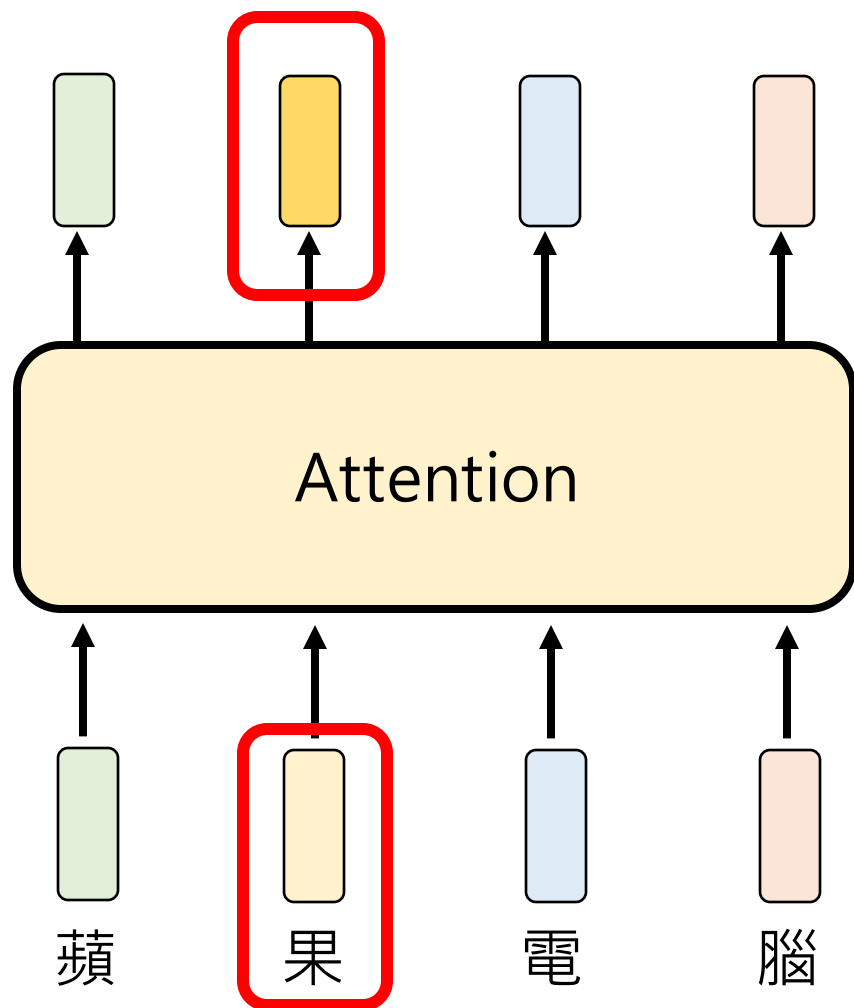


2. 理解每個 Token — 位置



3. Attention: 考慮上下文

Contextualized
Token Embedding



3. Attention: 考慮上下文

Attention Is All You Need

<https://arxiv.org/abs/1706.03762>

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

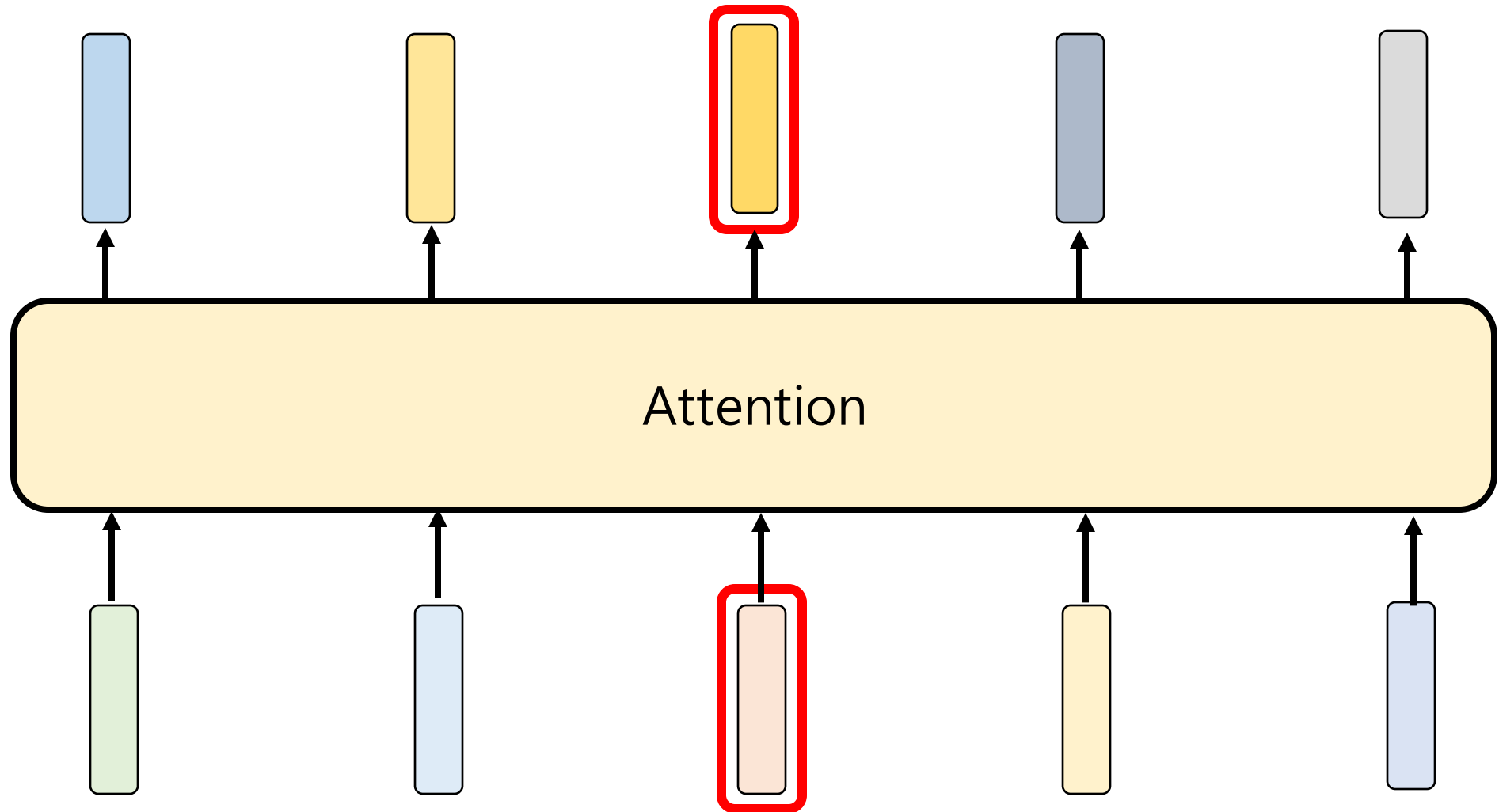
Aidan N. Gomez* †
University of Toronto
aidan@cs.toronto.edu

Łukasz Kaiser*
Google Brain
lukaszkaizer@google.com

Illia Polosukhin* ‡
illia.polosukhin@gmail.com

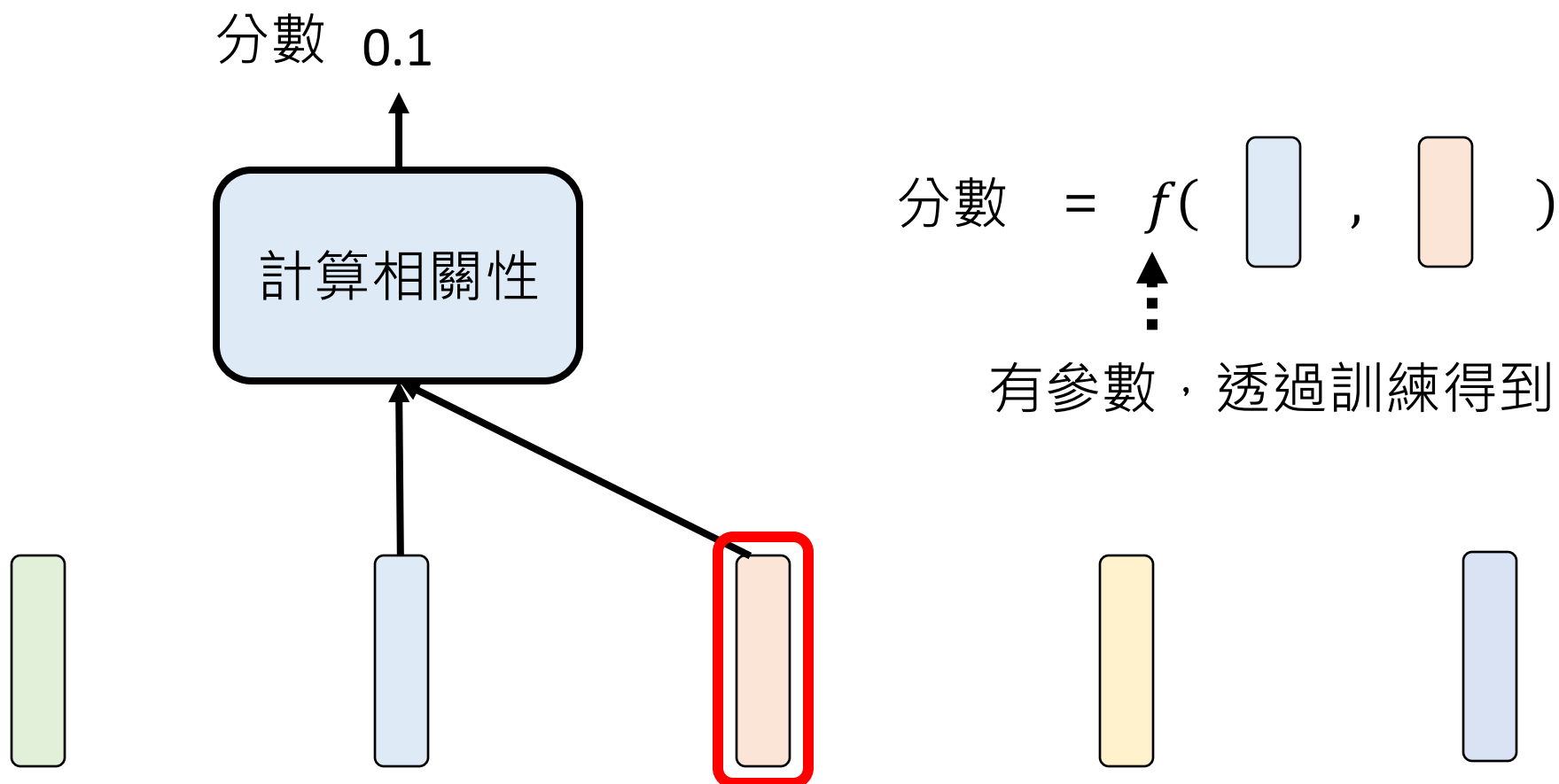
這篇文章主要的貢獻是發現不需要 Recurrent Neural Network (RNN) , 只需要 Attention 就夠了

3. Attention: 考慮上下文



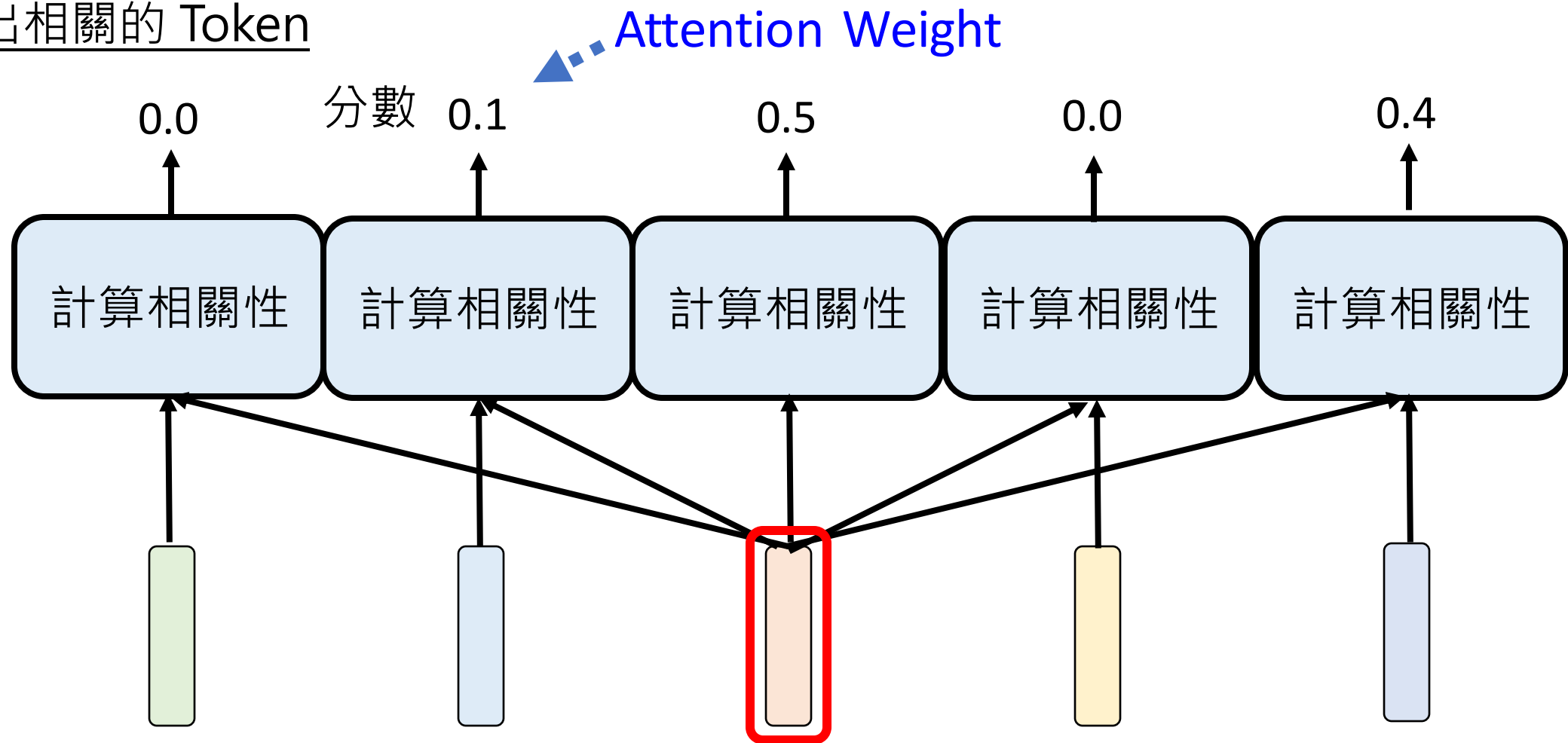
3. Attention: 考慮上下文

找出相關的 Token



3. Attention: 考慮上下文

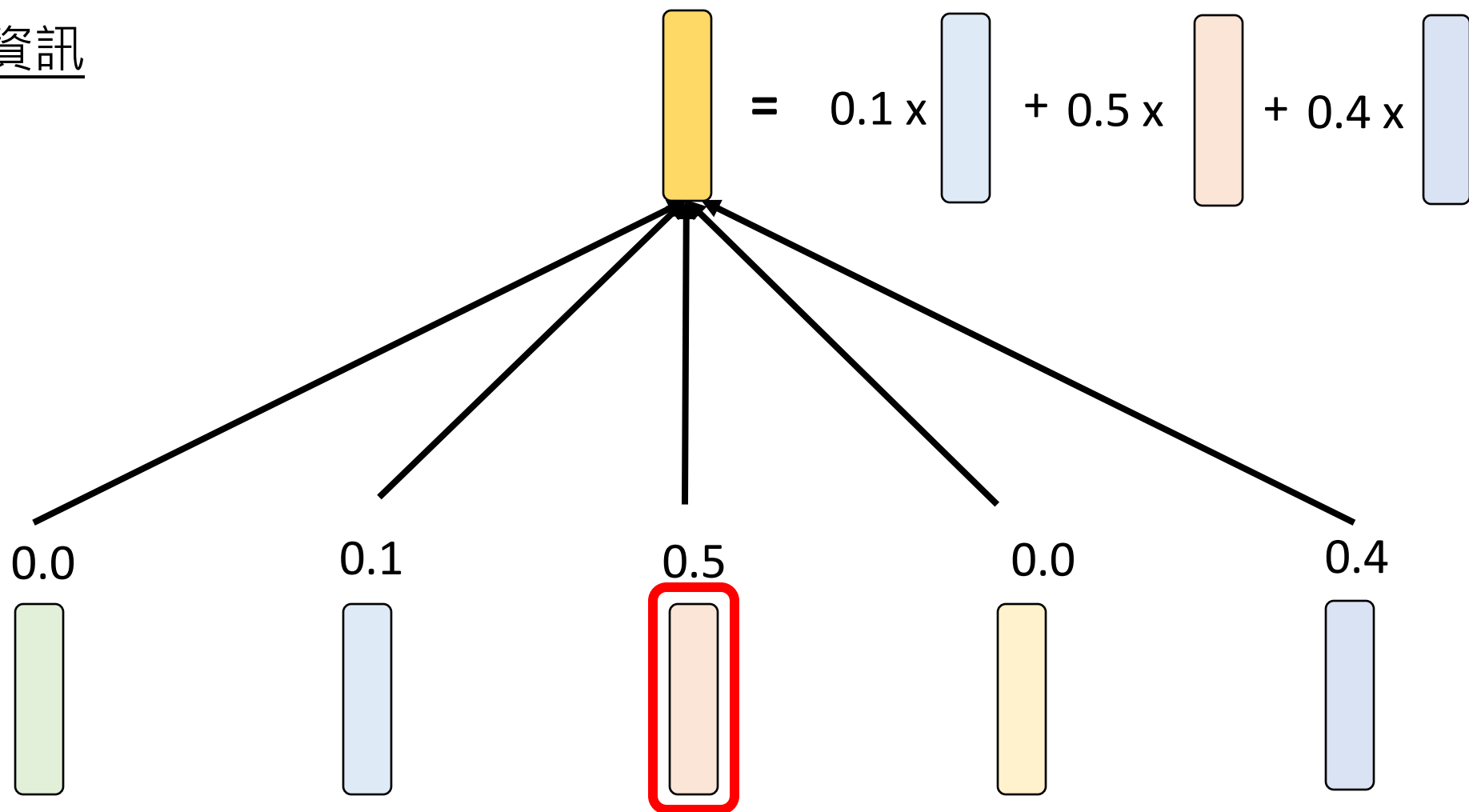
找出相關的 Token



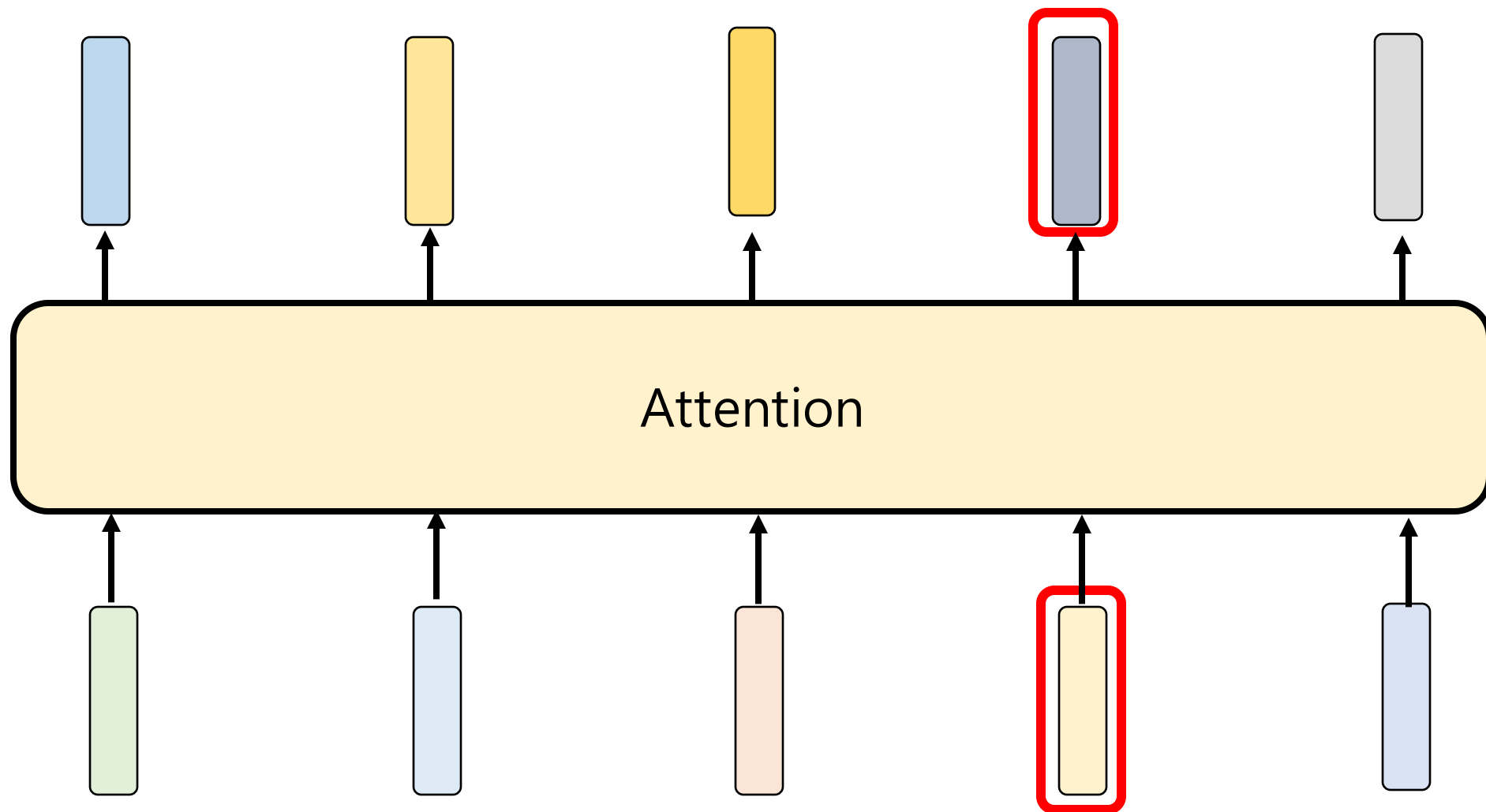
(此處為簡化的講法，事實上在相加前還要過一個 Linear Transform)

3. Attention: 考慮上下文

集合相關資訊

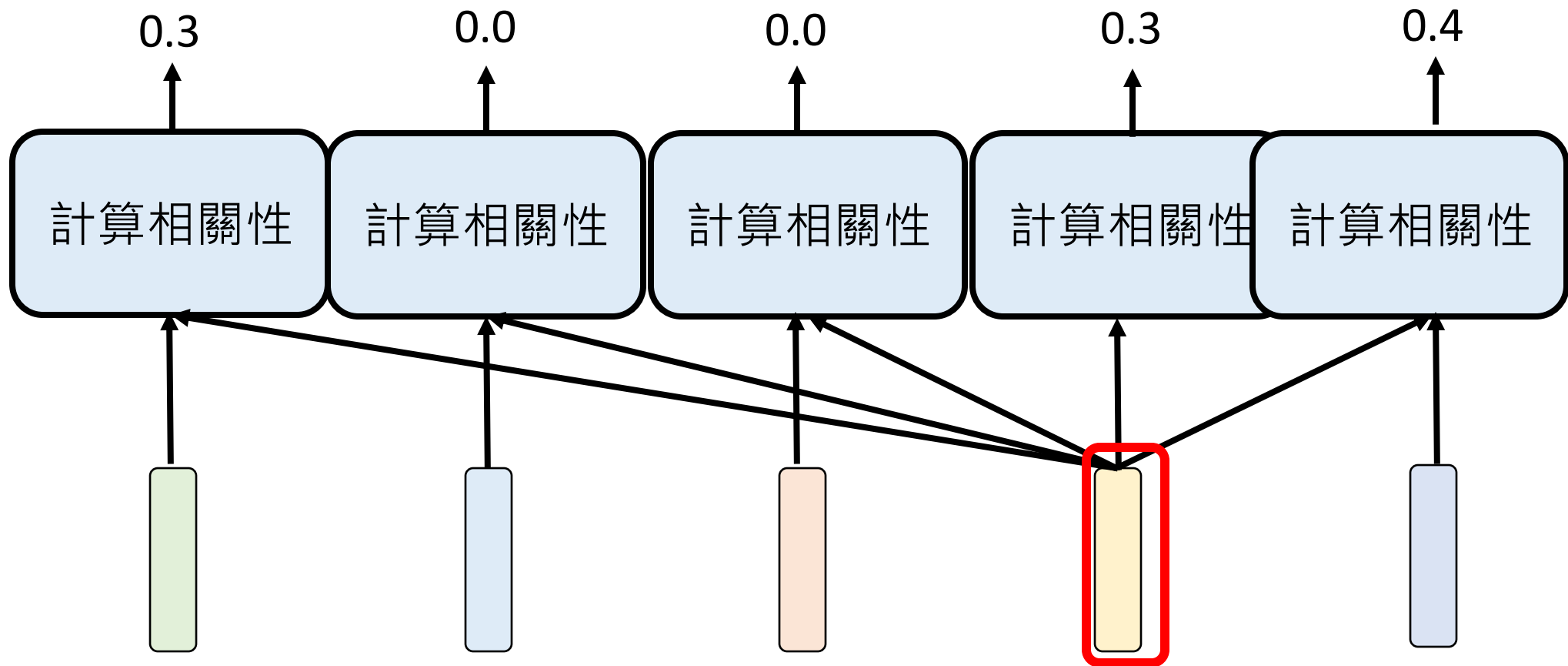


3. Attention: 考慮上下文



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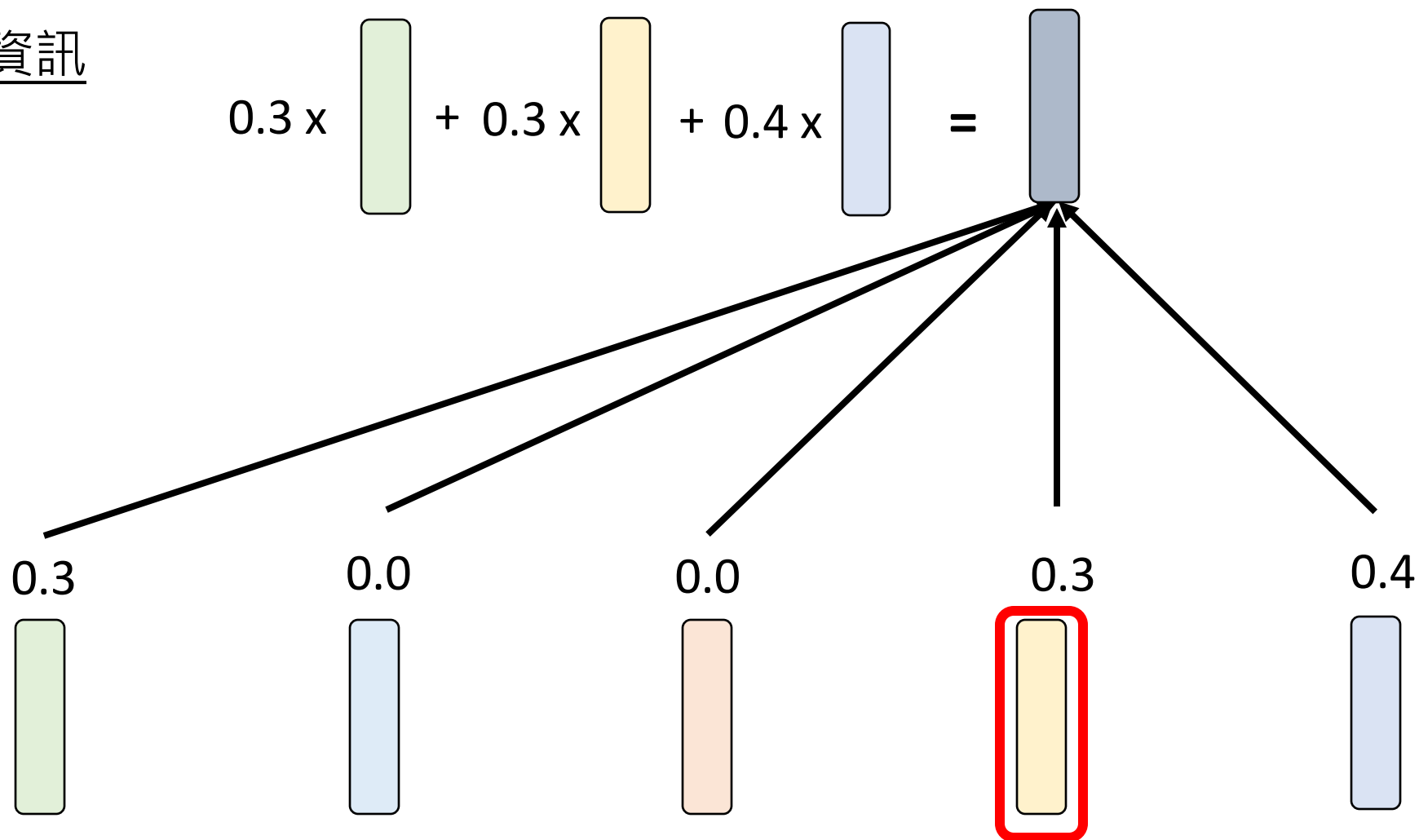
找出相關的 Token



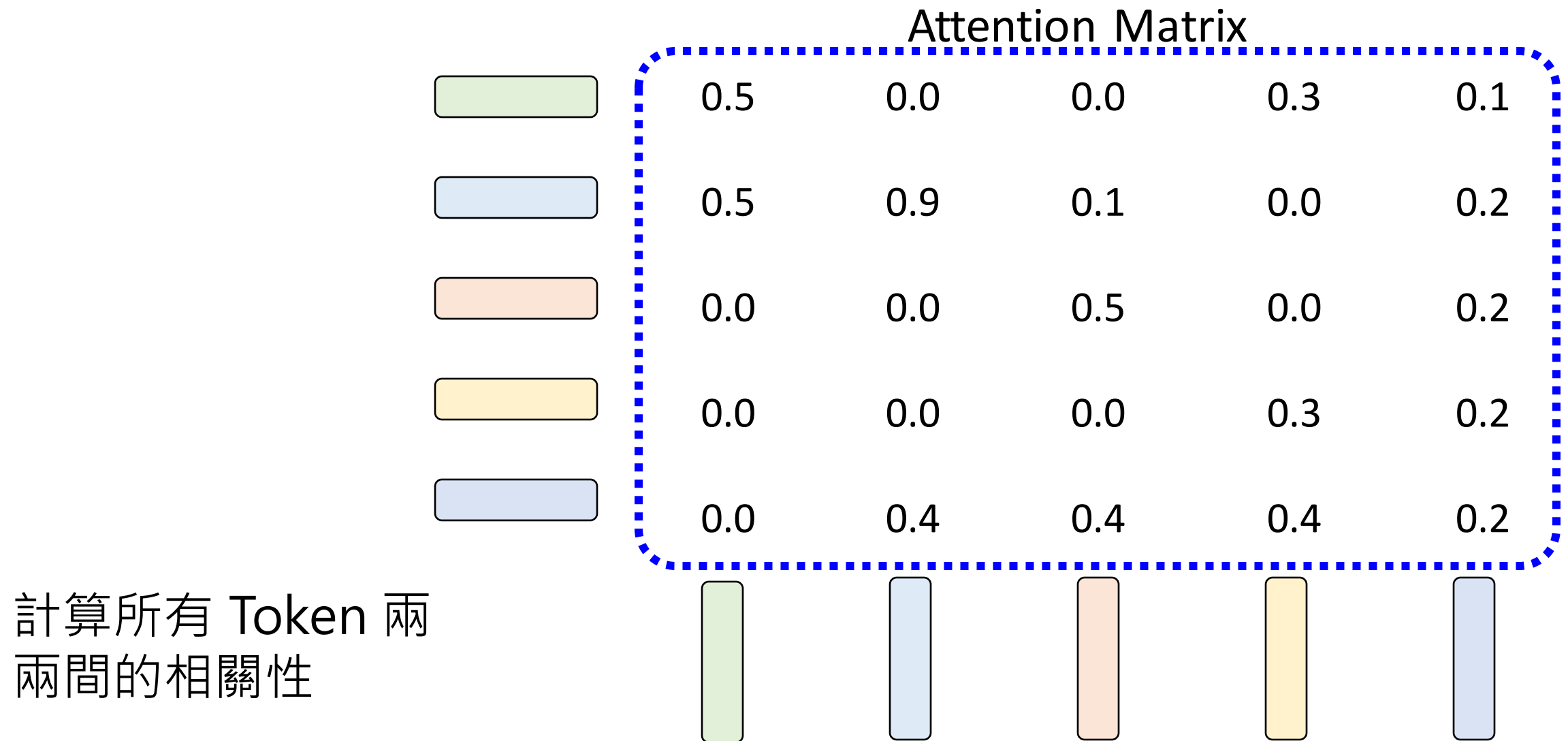
(此處為簡化的講法，事實上在相加前還要過一個 Linear Transformer)

3. Attention: 考慮上下文

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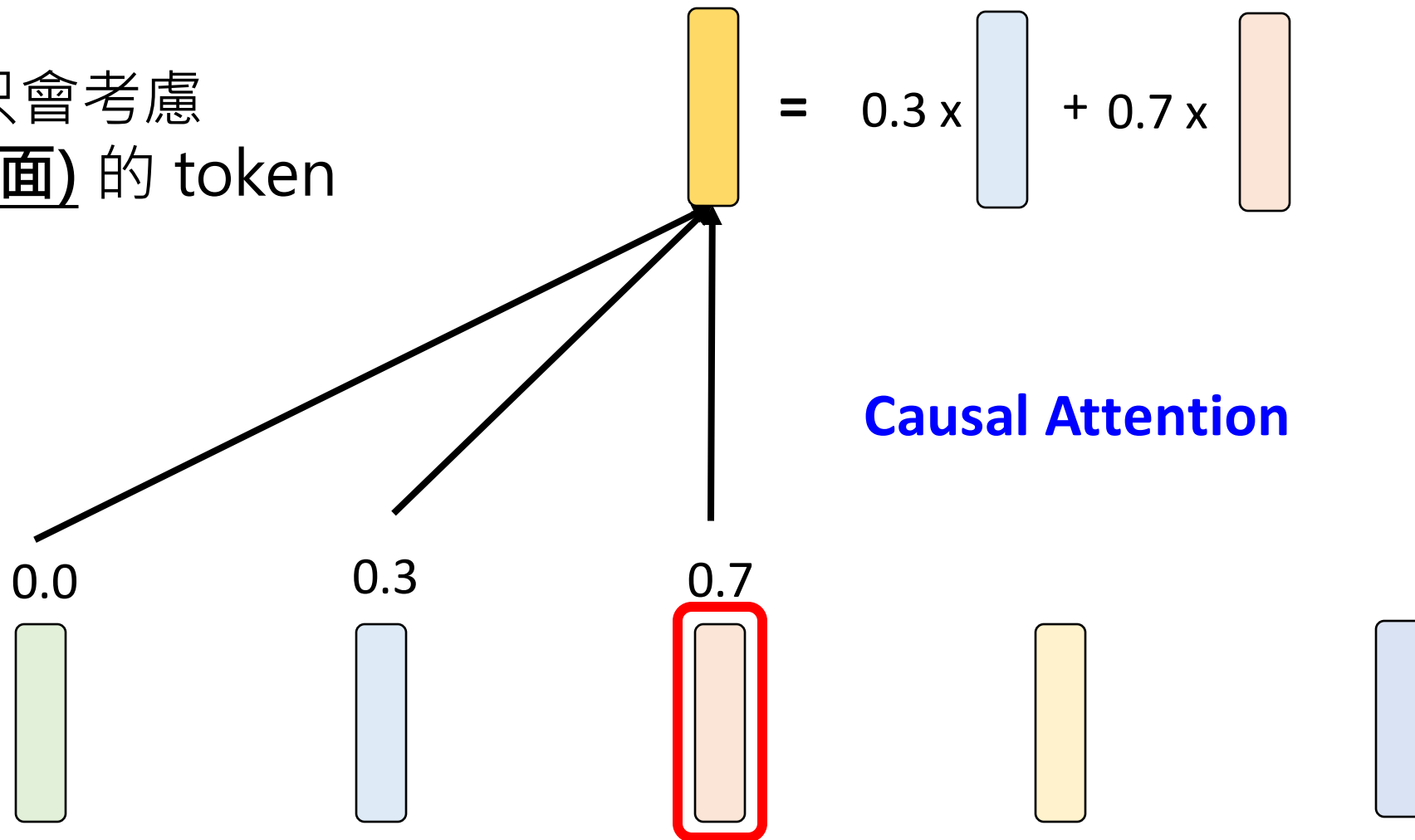


3. Attention: 考慮上下文



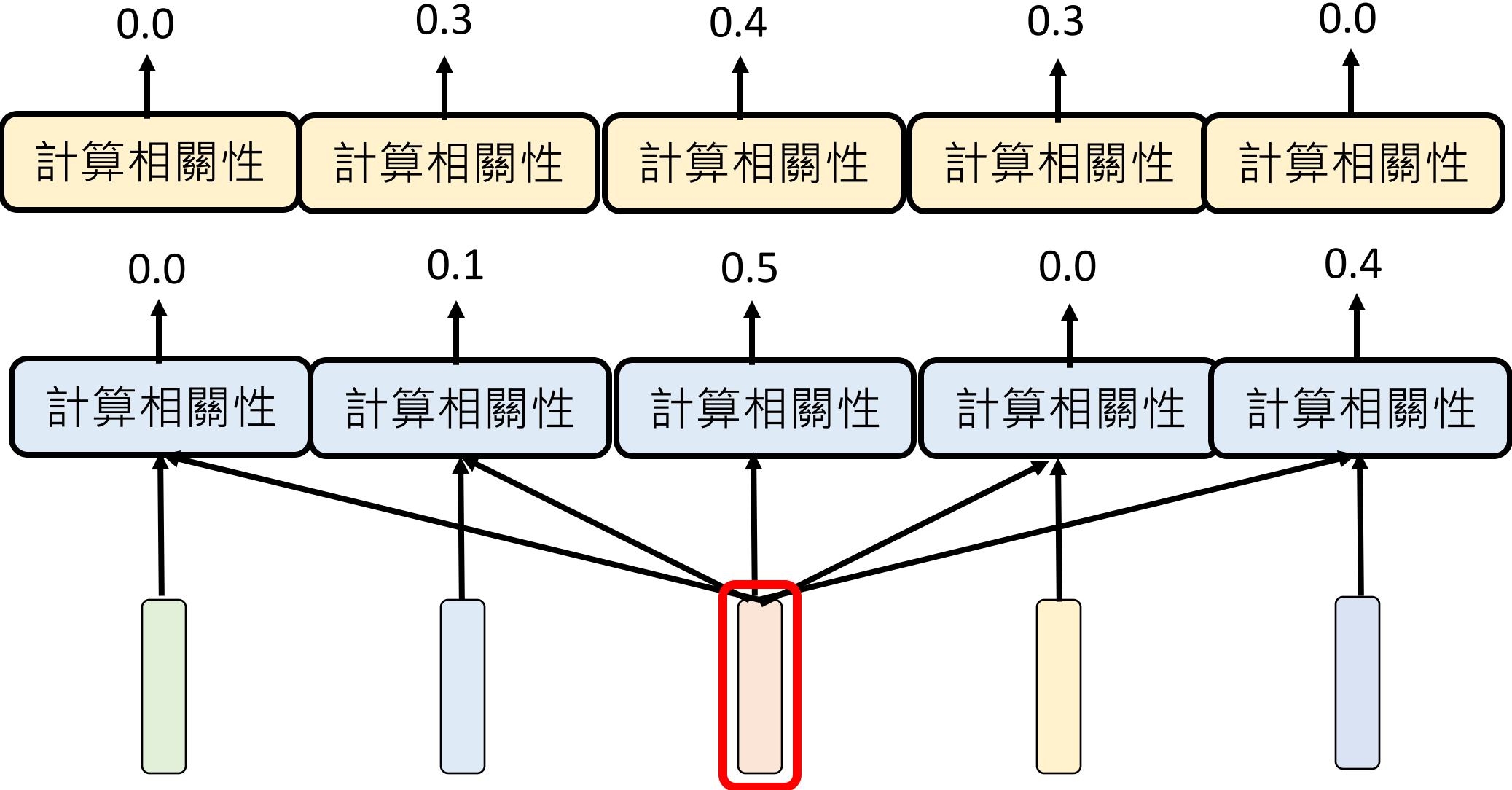
3. Attention: 考慮上下文

實作時只會考慮
左邊 (前面) 的 token

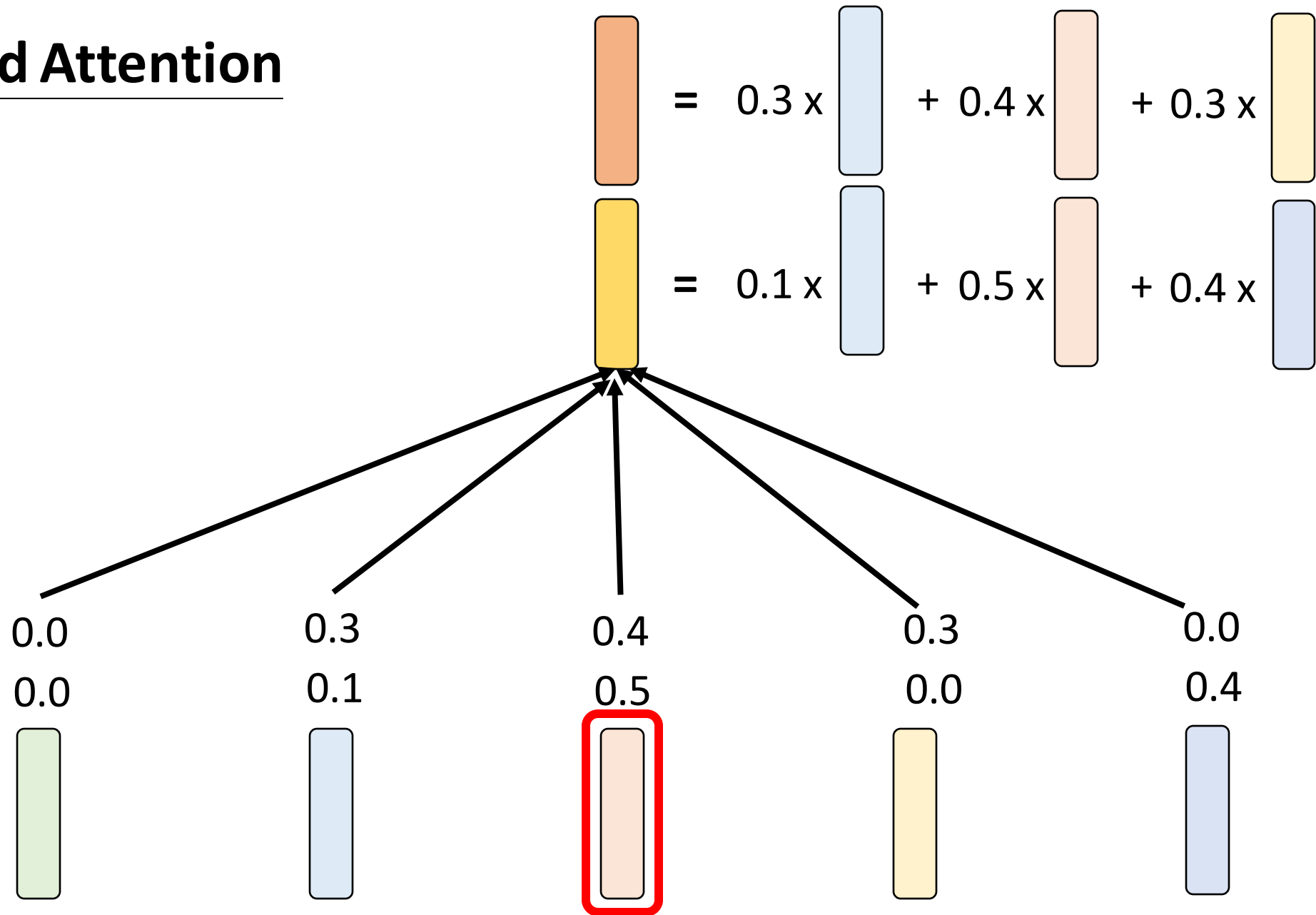


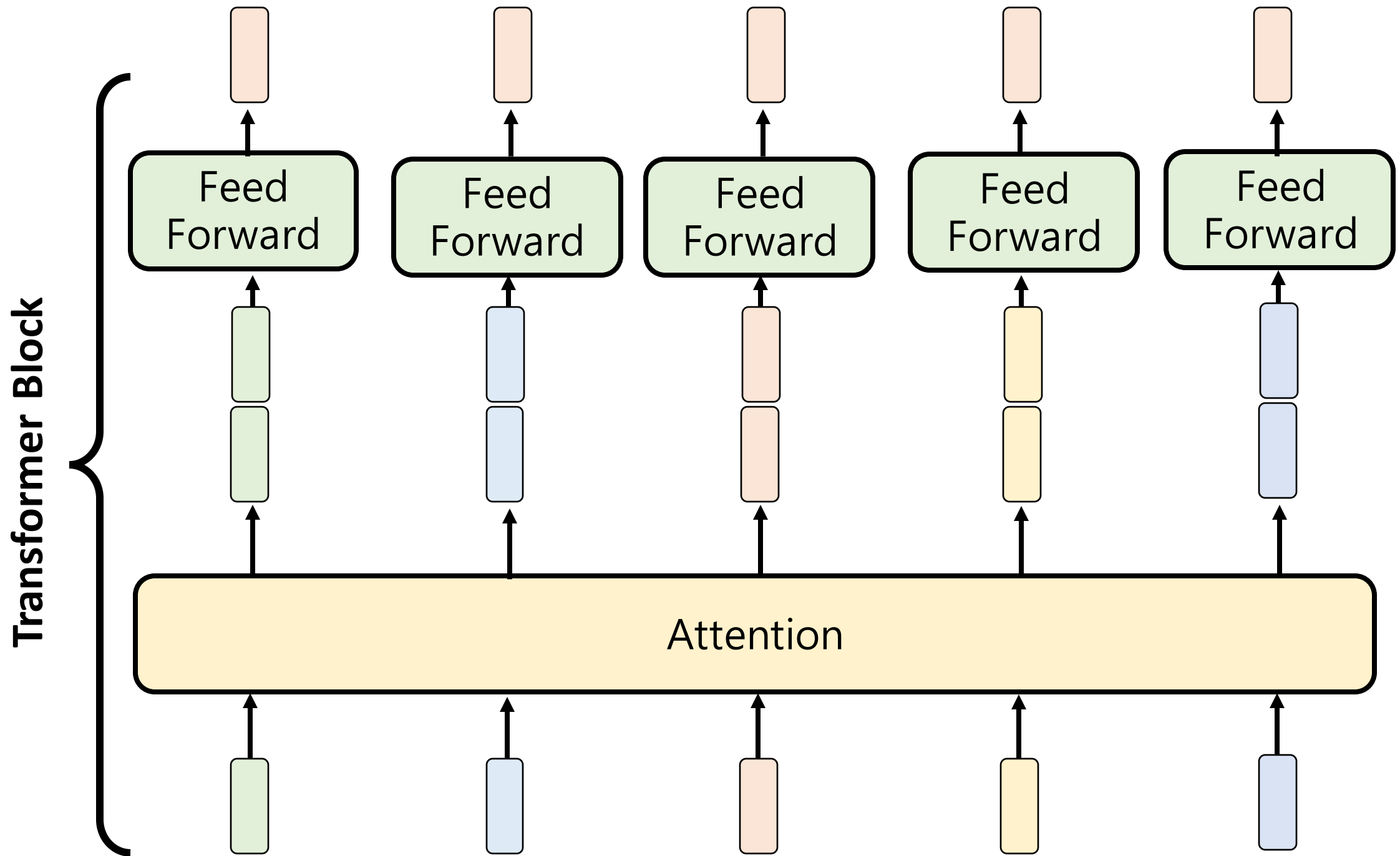
Multi-head Attention

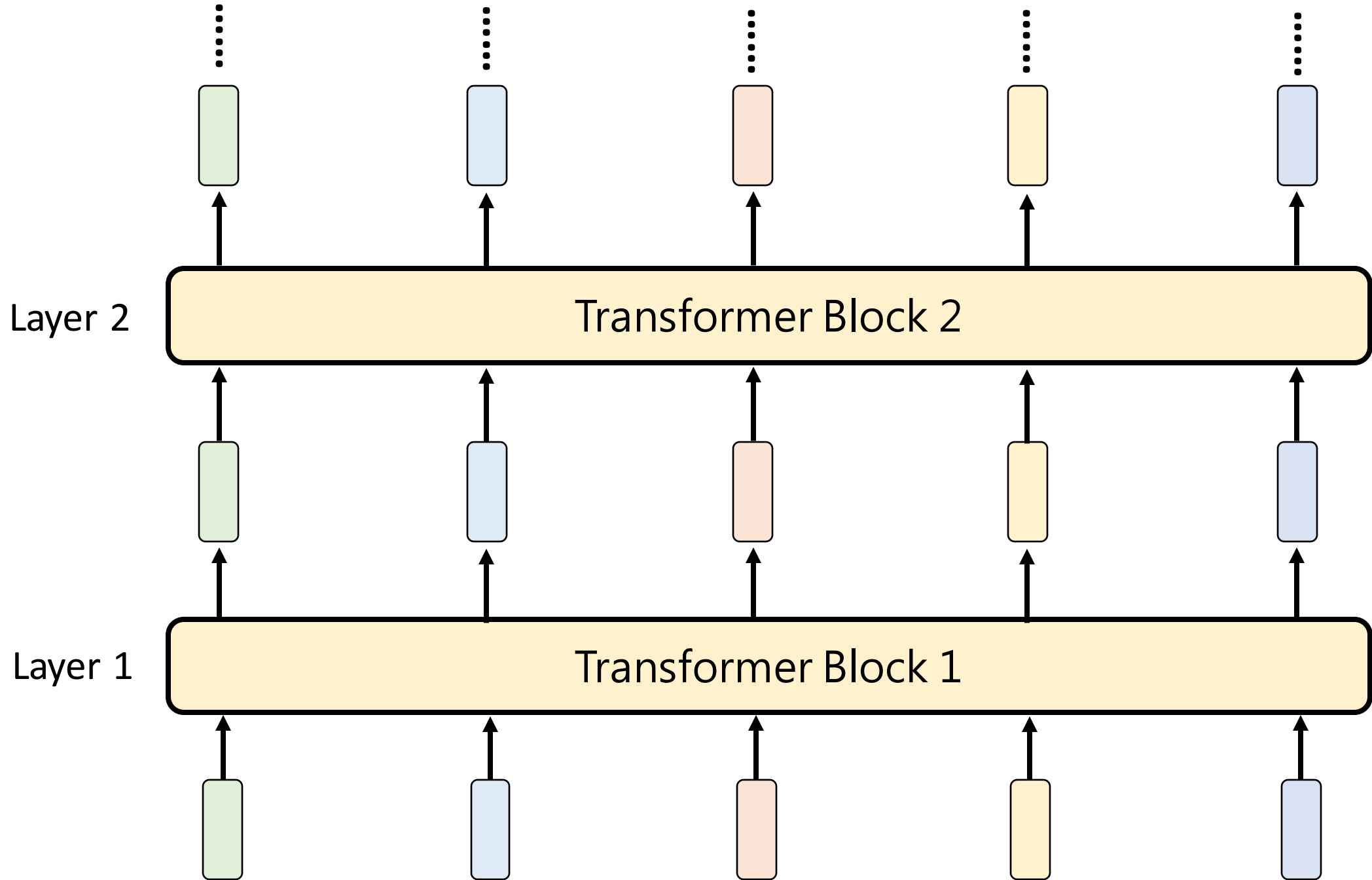
關聯性不只一種

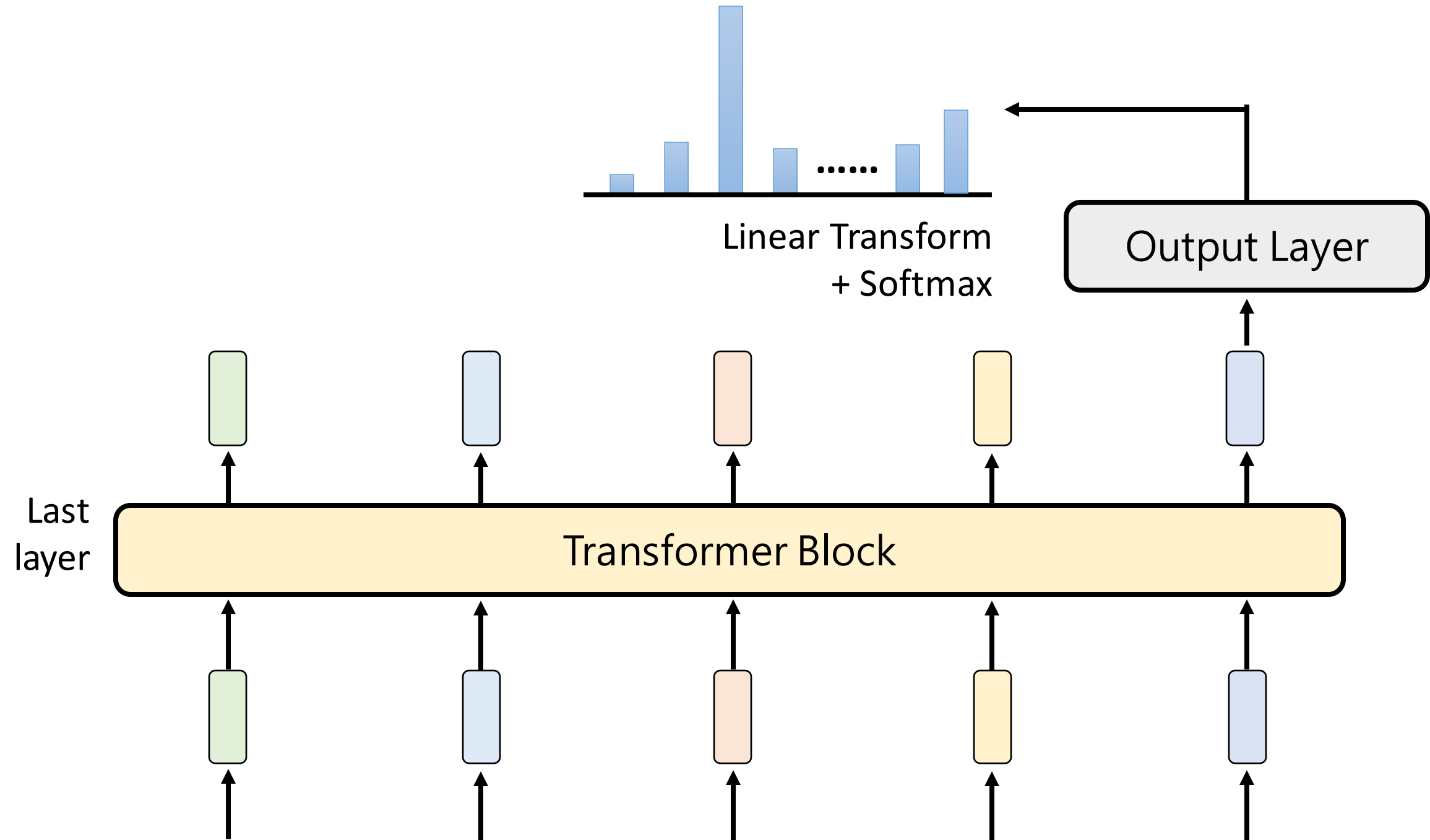


Multi-head Attention

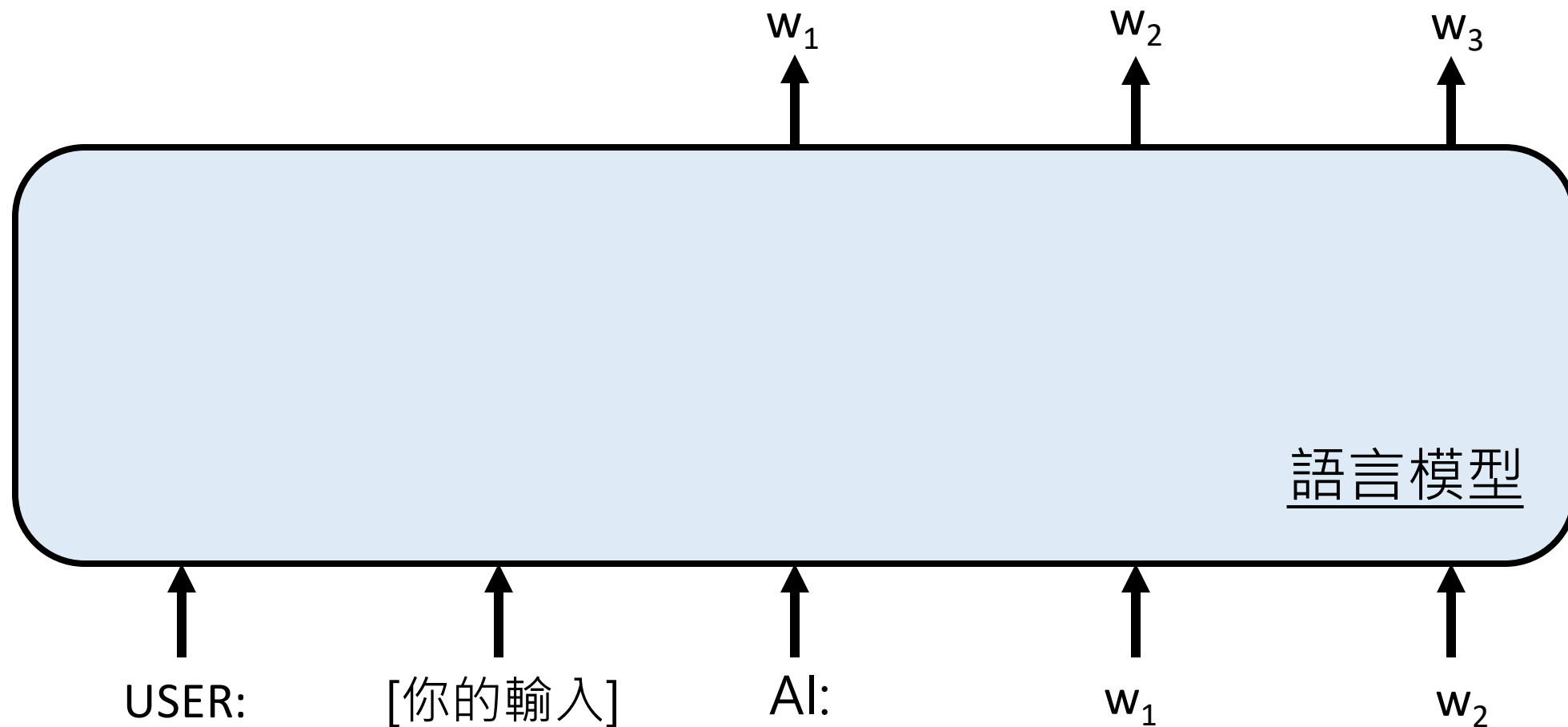




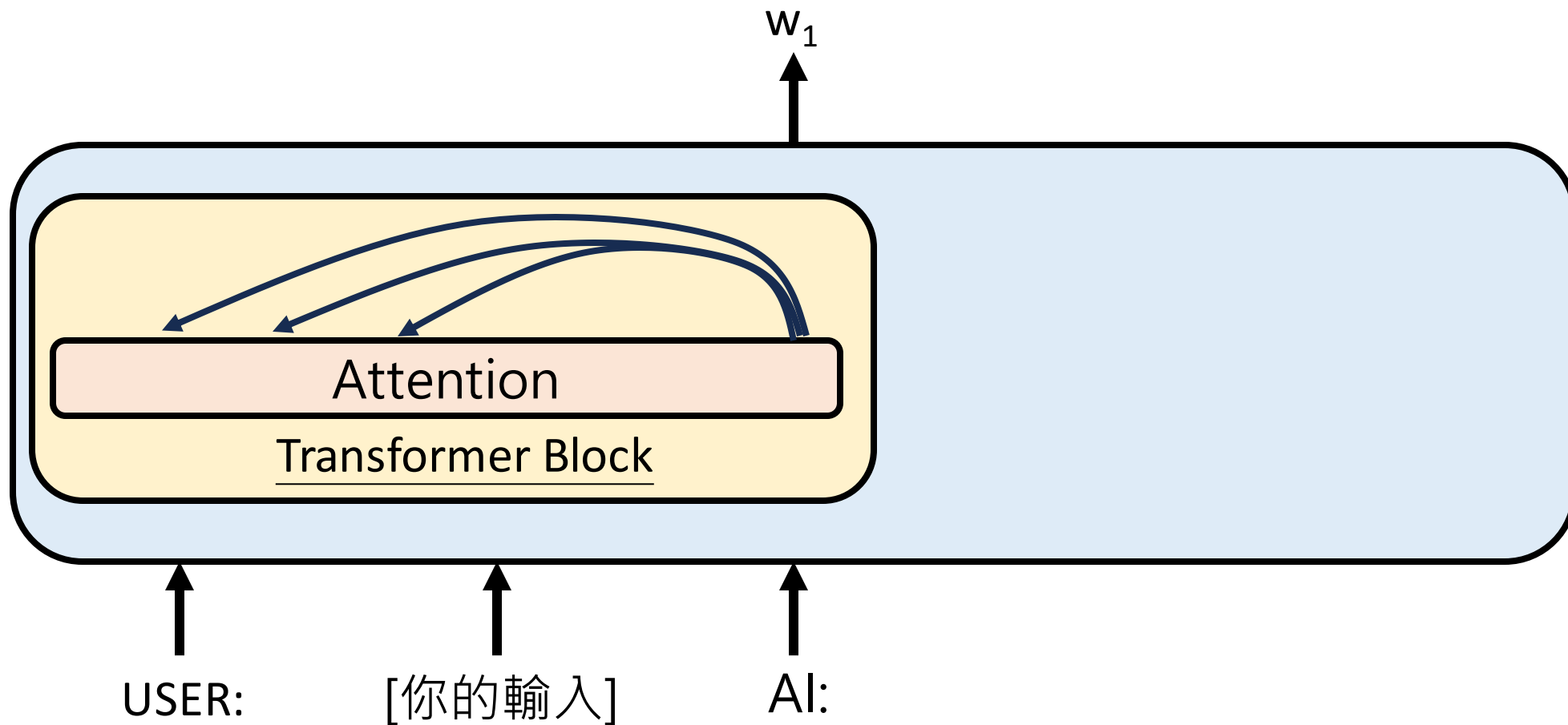




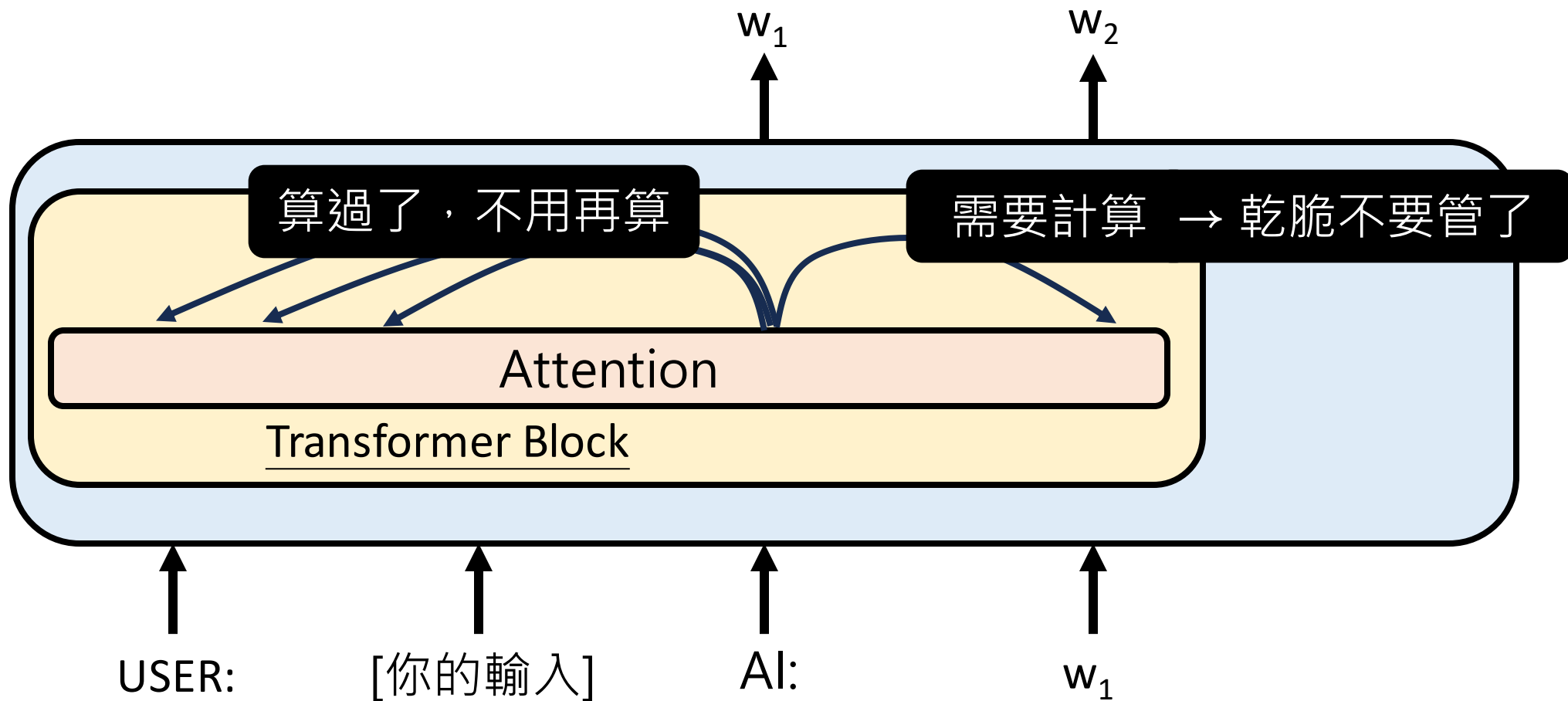
想想語言模型怎麼產生答案



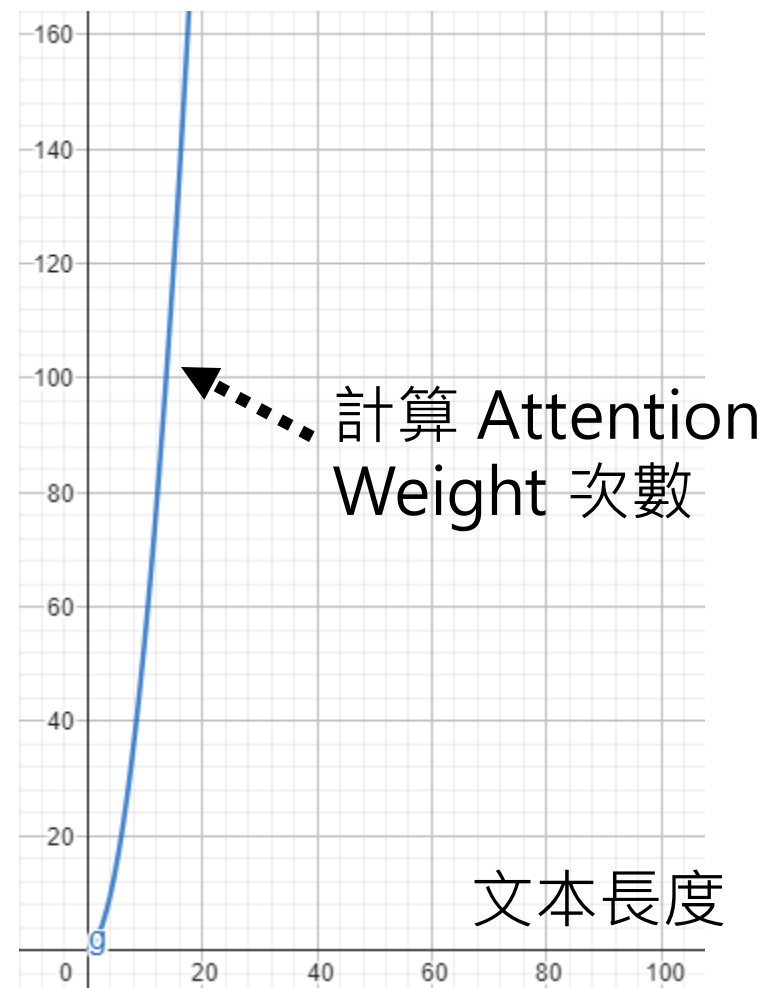
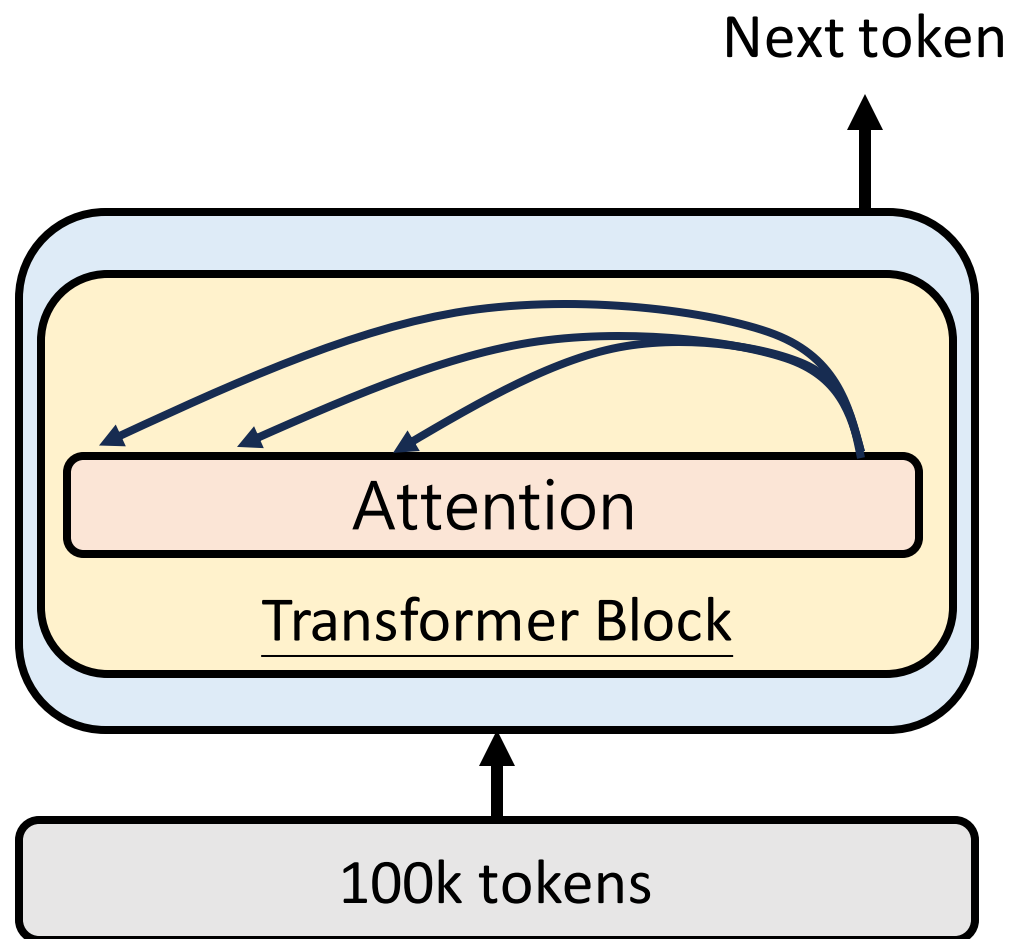
想想語言模型怎麼產生答案



想想語言模型怎麼產生答案



為什麼處理超長文本會是挑戰



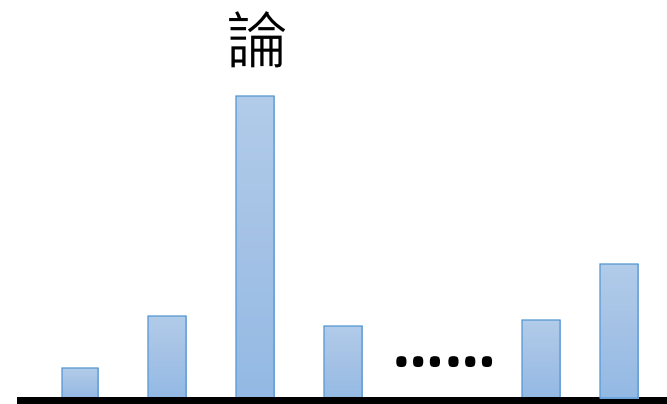
研究方向

- 如何快速進行 Attention
 - Overview: https://youtu.be/yHoAq1IT_og?si=2ZqqelyCYrt8l7DT
- 達成無限長度的 Attention?
 - <https://arxiv.org/abs/2309.17453>
 - <https://arxiv.org/abs/2404.07143>
- Train short, Test Long
 - ALiBi: <https://arxiv.org/abs/2108.12409>
 - Position Interpolation: <https://arxiv.org/abs/2306.15595> (RoPE: <https://arxiv.org/abs/2104.09864>)
- 其他類神經網路架構的可能性
 - Mamba: <https://arxiv.org/abs/2312.00752>
 - Jamba: <https://arxiv.org/abs/2403.19887>

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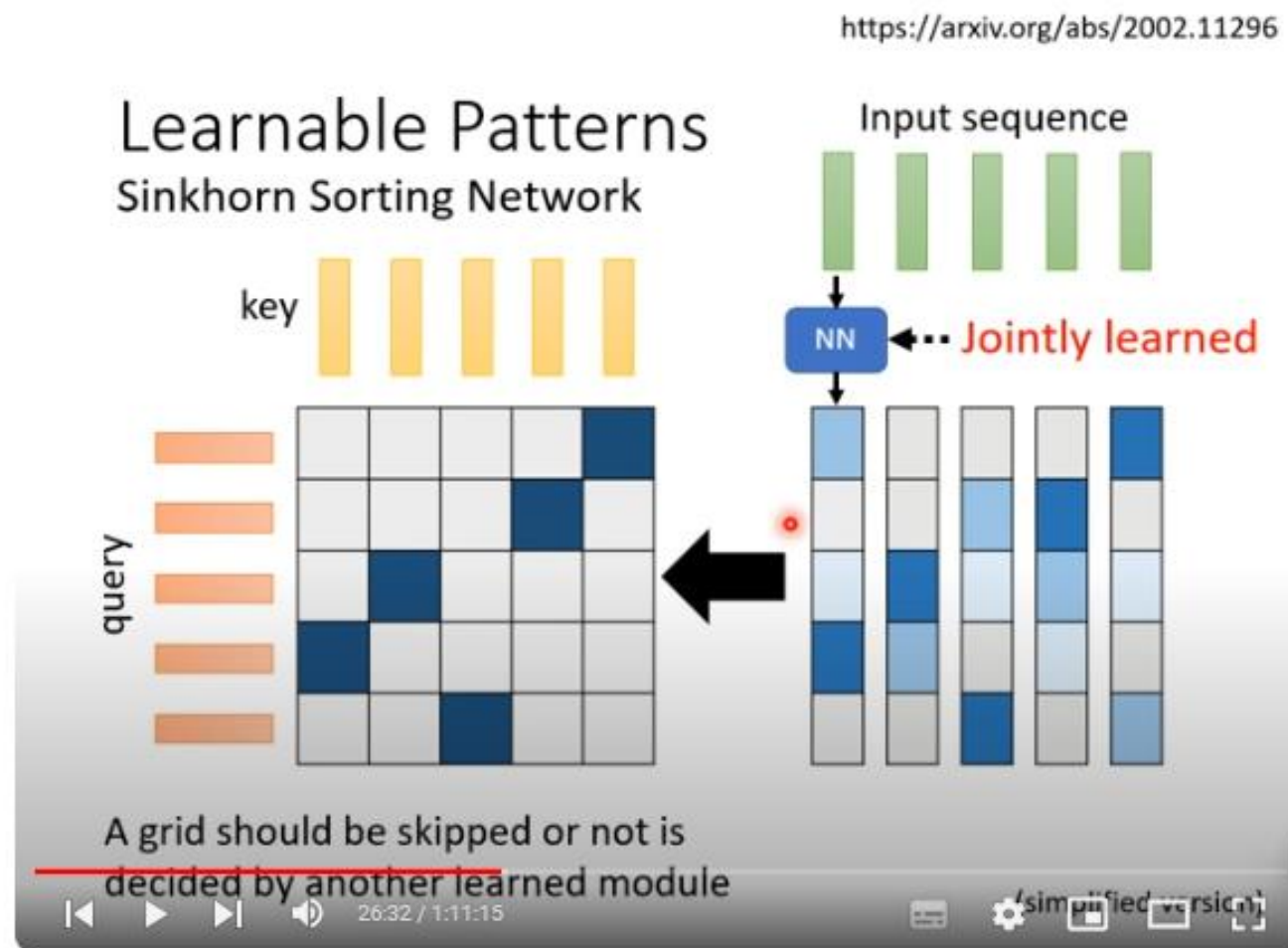
Transformer Block X N 反覆思考

Appendix

為什麼處理超長文本會是挑戰

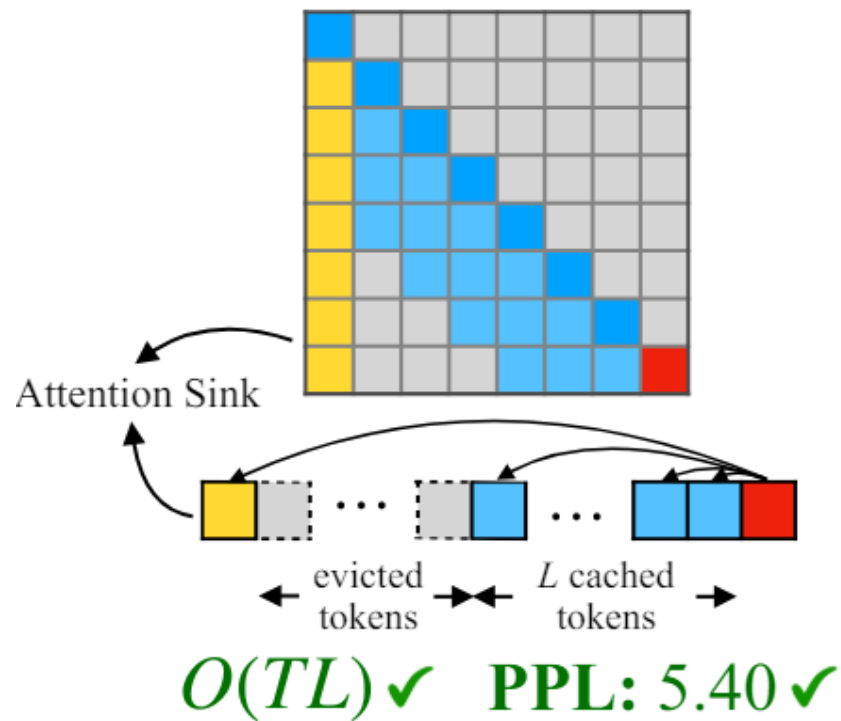
- 如何快速進行 Attention

https://youtu.be/yHoAq1IT_og?si=2ZqqelyCYrt8l7DT



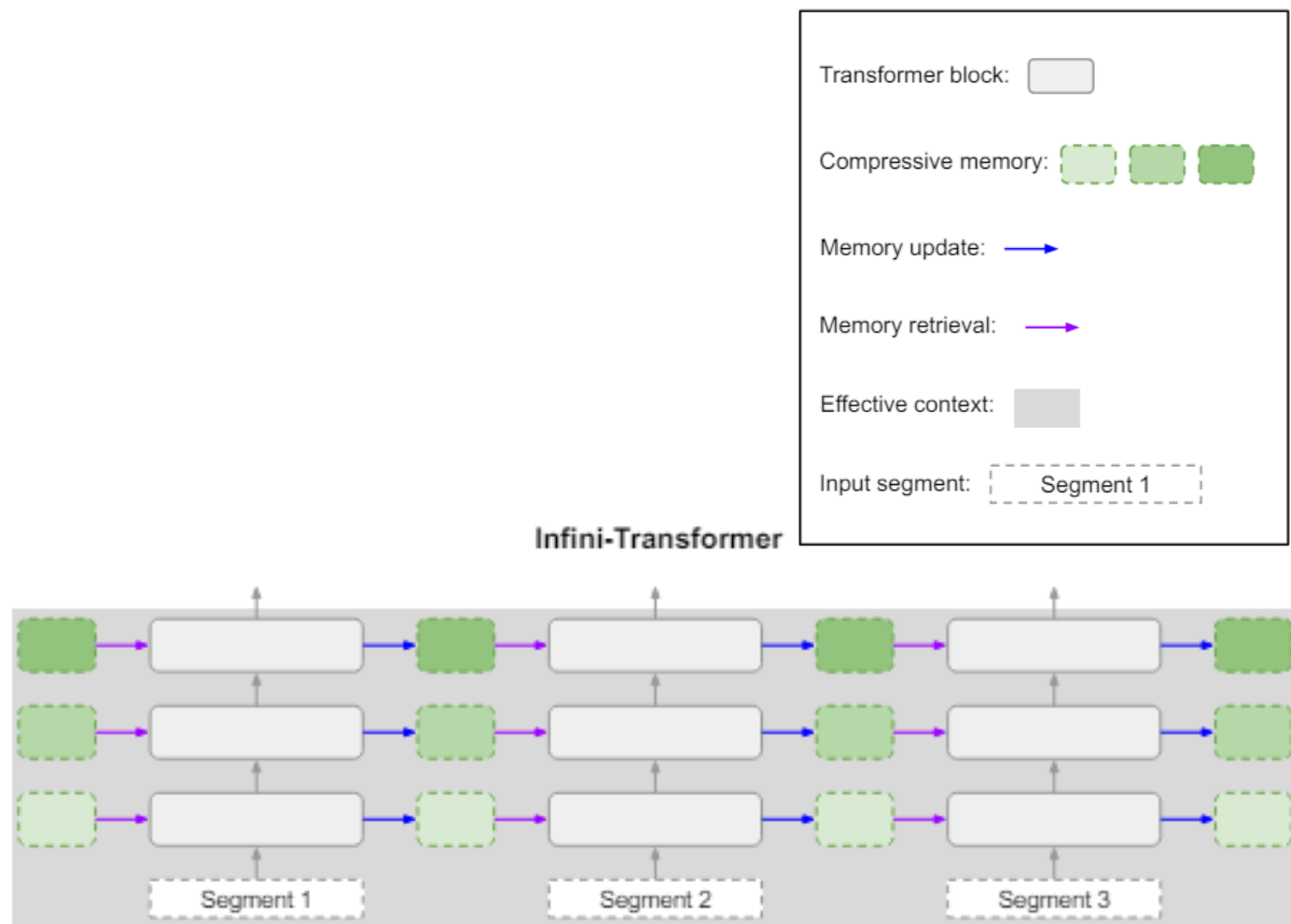
【機器學習 2022】各式各樣神奇的自注意力機制 (Self-attention) 變型

達到無限輸入？



Can perform efficient and stable language modeling on long texts.

<https://arxiv.org/abs/2309.17453>



<https://arxiv.org/abs/2404.07143>

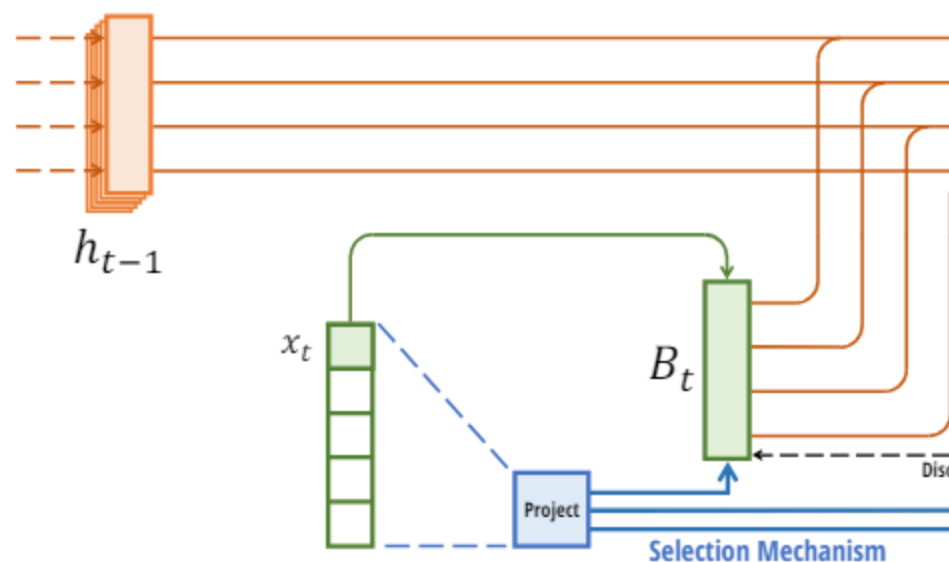
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<https://arxiv.org/abs/2312.00752>

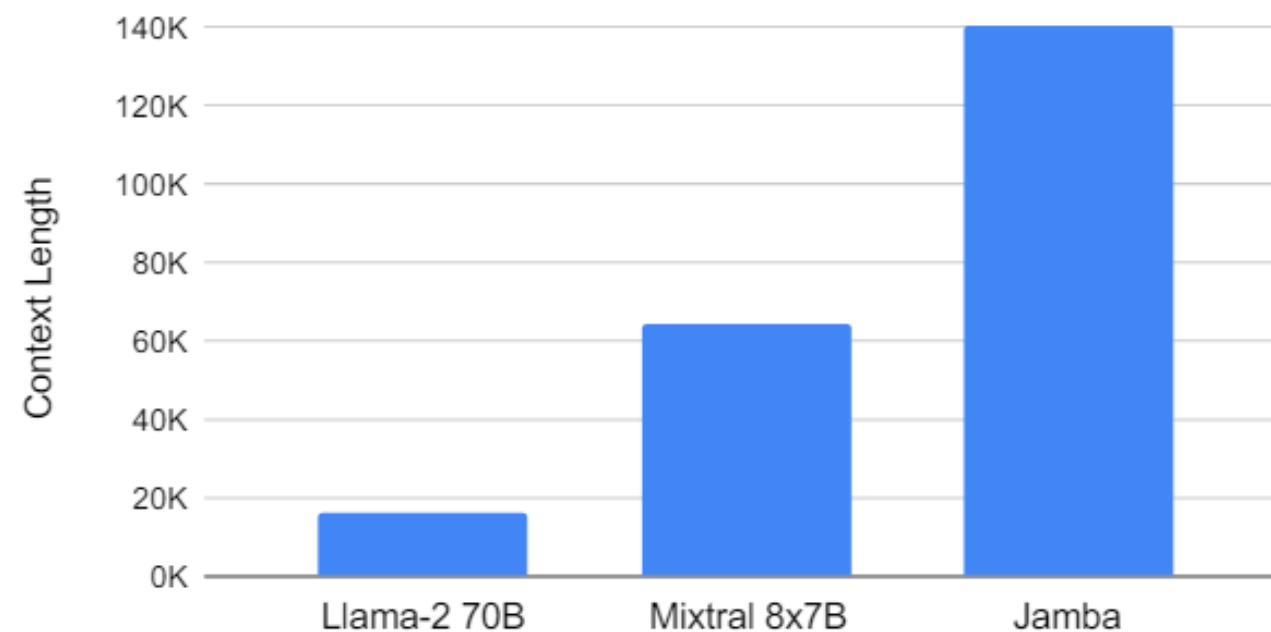
Jamba:

<https://arxiv.org/abs/2403.19887>



Mamba

Context length fitting a single 80GB A100 GPU



Jamba